MATH 419 MEASURE THEORY
LECTURE NOTES

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Abstract. These are the class notes for the course *Math 419 - Measure Theory* that I am teaching in the Fall 2026 semester at METU. (They are an evolution of class notes for the course *Math 319- Lebesgue Integral* which I have taught in Spring 2022 semester).

The notes are in constant evolution, they will be changed/corrected/enhanced almost daily. (Version date/time is below). All versions of these notes will contain errors/typos/non-sense. The author does not take responsibility for any harm that may be caused due to these.

I am using the books [1], [2] and [3] while preparing these notes.

These notes are for personal use and may not be sold!

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To whom it may concern

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1. Preliminaries

This section contains basic notions that will be used throughout these notes. Most of the content here will not be proven. One is advised to consult an undergraduate analysis book for a reference. This section will be enhanced as the notes expand.

1.1. **The Extended Real Number System.**

Since we will "measure" subsets of $\mathbb{R}^n$ and some subsets will have infinite measure (for example the whole real line has infinite "length"), it is useful to extend the real number system including infinity as an *extended real number*.

Definition 1.1. The *extended non-negative real axis* is $[0,\infty] = [0,\infty) \cup \{\infty\}$, in which the algebraic operations and order relations of $[0,\infty)$ are extended in the following way:

- For any $x \in [0,\infty], x + \infty = \infty + x = \infty$
- For any $x \in (0,\infty], x \cdot \infty = \infty \cdot x = \infty$
- $0 \cdot \infty = \infty \cdot 0 = 0$
- For any $x \in [0,\infty), x < \infty$

If (a_n) is a sequence in $[0,\infty]$ such that $\exists k, a_n = \infty, \forall n \geq k$, then we set $\lim a_n = \infty$.

For a sequence (a_n) in $[0,\infty]$, the series $\sum_{n=1}^{\infty} a_n$ has value ∞ if for some n, we have $a_n = \infty$. If all $a_n < \infty$, then the series either converges to a real number or it diverges to ∞. One can think the latter as "converging to ∞" in $[0,\infty]$. Hence, all series in $[0,\infty]$ are convergent.

Note that $\sum_{n=1}^{\infty} a_n$ is defined as $\lim_{N\to\infty} \sum_{n=1}^{N} a_n$ which is equal to

$$\sup_{\substack{F \subseteq \mathbb{N}^+ \\ F \text{ is finite}}} \left\{ \sum_{n \in F} a_n \right\}$$

So, for an arbitrary set $\{a_i \mid i \in I\}$ of elements of $[0,\infty]$, define

$$\sum_{i \in I} a_i = \sup_{\substack{F \subseteq I \\ F \text{ is finite}}} \left\{ \sum_{n \in F} a_n \right\}$$

We will make use of the following fact: (For a proof see [3])

Theorem 1.2. *Let* $\{a_{n,m} \mid (n,m) \in \mathbb{N}^+ \times \mathbb{N}^+\} \subseteq [0,\infty]$*, then*

$$\sum_{(n,m) \in \mathbb{N}^+ \times \mathbb{N}^+} a_{n,m} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{n,m} = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{n,m}$$

Definition 1.3. The *extended real numbers* is the set $\overline{\mathbb{R}} = \mathbb{R} \cup \{\mp\infty\} = [-\infty,\infty]$ in which the algebraic and order relations of $\mathbb{R}$ are extended in the following way:

- $-\infty < x < \infty, \forall x \in \mathbb{R}$
- $x \pm \infty = \mp\infty \ \forall x \in \mathbb{R}$, $\infty + \infty = \infty$, $-\infty - \infty = -\infty$
- $x \cdot (\mp\infty) = \mp\infty$, $\forall x \in (0,\infty]$ and for $x \cdot (\mp\infty) = \pm\infty$, $\forall x \in [-\infty, 0)$

- $0 \cdot (\mp\infty) = 0$

Note that we do not define $\infty + (-\infty) = \infty - \infty$.

Any subset of $\overline{\mathbb{R}}$ has a unique supremum (= least upper bound) and an infimum (=greatest lower bound). Th supremum is ∞ if and only if the set is not bounded above. Similarly, a set has infimum $-\infty$ if and only if it is not bounded below.

Definition 1.4. Let (a_n) be a sequence in $\overline{\mathbb{R}}$. Define

$$\liminf a_n = \sup_n \inf\{a_m \mid m \geq n\} = \lim_{n\to\infty} (\inf\{a_m \mid m \geq n\})$$
$$\limsup a_n = \inf_n \sup\{a_m \mid m \geq n\} = \lim_{n\to\infty} (\sup\{a_m \mid m \geq n\})$$

1.2. **The Heine-Borel Theorem.**

Recall the following fundamental notion from real analysis. We will use it in certain occasions.

Definition 1.5. A subset $C \subseteq \mathbb{R}^n$ is *compact*, if every open cover of C has a finite sub-cover, i.e., if $\{U_i \mid i \in I\}$ is a collection of open subsets of $\mathbb{R}^n$ with $C \subseteq \bigcup_{i\in I} U_i$, then there is a finite subset $J \subseteq I$ such that $C \subseteq \bigcup_{i\in J} U_i$.

Theorem 1.6. *(Heine-Borel Theorem) A subset $C \subseteq \mathbb{R}^n$ is compact if and only if it is closed and bounded.*

1.3. **Inverse images and continuity.**

If X and Y are sets and $f : X \to Y$ is a function, the *inverse image* of a subset $B \subseteq Y$ is the set

$$f^{-1}(B) = \{a \in X \mid f(x) \in B\}$$

Theorem 1.7. *If $\{B_i \mid i \in I\}$ is a collection of subsets of Y and $B \subseteq Y$, then*

(1) $f^{-1}(\bigcup_{i\in I} B_i) = \bigcup_{i\in I} f^{-1}(B_i)$
(2) $f^{-1}(\bigcap_{i\in I} B_i) = \bigcap_{i\in I} f^{-1}(B_i)$
(3) $f^{-1}(Y - B) = X - f^{-1}(B)$

Recall the following characterization of continuity:

Theorem 1.8. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a function. The following are equivalent:*

(1) *f is continuous*
(2) *$f^{-1}(U)$ is open for every open subset $U \subseteq \mathbb{R}$.*
(3) *$f^{-1}(a, b)$ is open for every $a, b \in \mathbb{R}$.*

2. Introduction

2.1. **What does it mean to measure?**

The basic "problem of measure theory" is to generalize the familiar notions of measure such as length, area, volume to more general subsets. Of course there are natural expectations from such a generalization.

Formally, having a way of "measuring" subsets of $\mathbb{R}^n$ means having a function (which we call a *measure)*

$$\mu : \mathcal{P}(\mathbb{R}^n) \to [0, \infty]$$

Clearly we want to assign to "elementary sets" (such as intervals in $\mathbb{R}$, rectangles in $\mathbb{R}^2$, boxes in $\mathbb{R}^3$ etc.) their natural measure (i.e., length, area, volume etc.). It is then (kind of but not exactly!) clear what measure one should assign to sets which are finite unions of such elementary sets. The (not so small) problem here is that such a set may be expressed as a union of the elementary sets in many different ways and one has to show that the measure does not depend on the particular expression.

Then comes a more serious problem: How to measure sets which are not of this form? For example, how to measure the area of the unit disk in $\mathbb{R}^2$? One should realize that this is related to the Riemann Integral that you must have seen in earlier courses. But, as we will see in later sections, this method works fine for various situations but has severe shortcomings!

So, what *natural* properties do we want from a measure: $\mu : \mathcal{P}(\mathbb{R}^n) \to [0, \infty]$?

Here are some such properties:

(1) If $E_1, E_2, \dots$ are disjoint sets, then

$$\mu(E_1 \cup E_2 \cup \cdots) = \mu(E_1) + \mu(E_2) + \dots$$

(This is called *countable additivity*)

(2) If E is obtained from F by translations, rotations or reflections, then

$$\mu(E) = \mu(F)$$

(This is called *invariance*)

(3) If $E = [0, 1] \times [0, 1] \times \cdots \times [0, 1]$ (i.e, E is the unit interval, square, cube when $n = 1, 2, 3$ respectively), then $\mu(E) = 1$.

The (surprising) fact is that no such function exists! Let us show this for $n = 1$ (a similar argument works for $n > 1$). This argument is due to *Giuseppe Vitali* (1905):

2.2. **One cannot measure all sets.**

(1) Consider the equivalence relation $\sim$ defined on $[0, 1]$ as

$$x \sim y \text{ if } x - y \in \mathbb{Q}$$

(2) Let $V \subseteq [0,1]$ be a set containing exactly one element from each equivalence class.[1]
(3) For a real number $r \in \mathbb{R}$, let $V + r = \{v + r \mid v \in V\}$, i.e, *translate* V by r.
Let $\mathbb{Q} \cap [-1,1] = \{q_1, q_2, \ldots\}$ and let $V_i = V + q_i$. Note that $V_i \subseteq [-1,2]$ and if $i \neq j$ then $V_i \cap V_j = \varnothing$.
(If $z \in V_i \cap V_j$, then $z = x + q_i = y + q_j$ for some $x, y \in V$. This implies $x - y \in Q$ and hence $x = y$. Which in turn implies $q_i = q_j$).
(4) Given $y \in [0,1]$, $y \sim x$ for some $x \in V$. So, $y - x \in \mathbb{Q} \cap [-1,1]$. Thus, $y - x = q_j$ for some j. It follows that $y = x + q_j \in V_j$. This shows that $[0,1] \subseteq \bigcup_{i=1}^{\infty} V_i \subseteq [-1,2]$
(5) Now suppose that a function μ exists as in the previous section. Then for any i, $\mu(V_i) = \mu(V + q_i) = \mu(V)$, so
(6) $\mu(\bigcup_{i=1}^{\infty} V_i) = \sum_{i=1}^{\infty} \mu(V_i) = \sum_{i=1}^{\infty} \mu(V)$.
(7) But from the inclusions of item (4) we must have $1 \leq \sum_{i=1}^{\infty} \mu(V) \leq 3$.

This is a contradiction since either $\mu(V) > 0$, in this case $\sum_{i=1}^{\infty} \mu(V) = \infty$, or $\mu(V) = 0$, in this case $\sum_{i=1}^{\infty} \mu(V) = 0$.

(Let us call this set V a *Vitali set*.)

How to "remedy" this situation? One possible suggestion is to change the requirement (1) of a measure to hold only for *finite* collections only, i.e.:

If $E_1, E_2, \ldots, E_n$ are disjoint sets, then $\mu(E_1 \cup E_2 \cup \cdots \cup E_n) = \mu(E_1) + \mu(E_2) + \ldots + \mu(E_n)$

(This is called *finite additivity*)

Unfortunately, this does not solve the problem entirely:

2.3. **The Banach-Tarski Theorem (Paradox).**

The following was proven by Banach and Tarski in 1924[2]:

Theorem 2.1. [3] *Let $B = \{(x,y,z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq 1\}$ be the unit ball in $\mathbb{R}^3$. B can be decomposed into finitely many pieces, which after translations and rotations can be reassembled into two disjoint copies of B.*

More precisely, there are disjoint subsets $A_1, A_2 \ldots, A_n, B_1, B_2, \ldots, B_m \subseteq B$ and

$T_1, T_2, \ldots, T_n, S_1, S_2, \ldots, S_m$ which are translations or rotations in $\mathbb{R}^3$ such that

$$B = A_1 \cup A_2 \cup \cdots \cup A_n \cup B_1 \cup B_2 \cup \cdots \cup B_m$$

with

$$B = T_1(A_1) \cup T_2(A_2) \cup \cdots \cup T_n(A_n) \text{ and } B = S_1(B_1) \cup S_2(B_2) \cup \cdots \cup S_m(B_m)$$

[1] This is "easy", one just picks one element from each equivalence class and forms V. This is related to the *axiom of choice*! Maybe we will talk about this later?

[2] Once again, this relies on the *axiom of choice*.

[3] Click here to watch a video outlining a proof of this.

(In words, out of B, one can make two disjoint copies of B. In fact a general theorem of Banach and Tarski states that in $\mathbb{R}^n, n \geq 3$ any two bounded subsets with non-empty interiors can be transformed into one other in a similar fashion).

This theorem prohibits the existence of a finitely additive measure on $\mathbb{R}^3$ which assigns positive measure to B. (Note that the "problem" with the conclusion of the theorem arises if one wants to measure the pieces of the decomposition)

The moral of this story is that the sets in the above decomposition are "pathological" (or "fractal") and one should avoid talking about their volume. That is:

Some subsets of $\mathbb{R}^n$ are so bizarre, we should not try to measure them!

In other words, the solution to the problem is to drop the requirement to measure all subsets and rather restrict attention to a class of "nice" subset which will be called *measurable sets*. Examples of *non-measurable* sets include V of subsection 2.2 and the pieces of the unit ball in the previous section.

Of course, the aim is to be able to measure as many sets as possible, while keeping the measure as nice as possible.

2.4. **Riemann vs. Lebesgue.** [1]

Recall the following definition of the Riemann integral of a bounded function:

If $[a, b]$ is an interval in $\mathbb{R}$, a *partition* of $[a, b]$ is a sequence of points $P = \{t_0, t_1, \ldots, t_n\}$ such that $a = t_0 < t_1 < \cdots < t_{n-1} < t_n = b$.

Let $f : [a, b] \to \mathbb{R}$ be a bounded function and $P = \{t_0, t_1, \ldots, t_n\}$ a partition of $[a, b]$. For $i = 1, \ldots, n$ let $M_i = \sup\{f(t) \mid t \in [t_{i-1}, t_i]\}$ and $m_i = \inf\{f(t) \mid t \in [t_{i-1}, t_i]\}$. We define the upper and lower Riemann sums of f with respect to P by

$$U(f, P) = \sum_{i=1}^{n} M_i(t_i - t_{i-1}) \text{ and } L(f, P) = \sum_{i=1}^{n} m_i(t_i - t_{i-1}).$$

Let

$$U(f) = \inf\{U(f, P) \mid P \text{ is a partition of } [a, b]\}$$
$$L(f) = \sup\{U(f, P) \mid P \text{ is a partition of } [a, b]\}$$

A bounded function $f : [a, b] \to \mathbb{R}$ is *Riemann integrable* if $U(f) = L(f)$ in which case the common value is the *Riemann integral* of f on $[a, b]$ denoted by $\int_a^b f(x)\, dx$.

The class of Riemann integrable functions does not have some desirable properties.

For example the following function (usually called the *Dirichlet function*) is not Riemann integrable (show this as an exercise!):

$$f(x) = \begin{cases} 1 & x \in [0, 1] \cap \mathbb{Q} \\ 0 & x \in [0, 1] - \mathbb{Q} \end{cases}$$

[1] Pronounced as "Löbek".

("Normally" one would "want" this function to have zero integral.)

Another, undesired point is that the set of Riemann integrable functions is not *complete* (in the sense of metric spaces):

Since the rational numbers are countable, let $[0,1] \cap \mathbb{Q} = \{r_1, r_2, r_3, \ldots\}$ be an enumeration of $[0,1] \cap \mathbb{Q}$. For each $n \geq 1$ define

$$f_n(x) = \begin{cases} 1 & x \in \{r_1, r_2, \cdots, r_n\} \\ 0 & x \in [0,1] - \{r_1, r_2, \cdots, r_n\} \end{cases}$$

As exercise show that each f_n is Riemann integrable and f_n converge (pointwise) to the Diricihlet function. Thus, the space of Riemann integrable functions has "holes" (= is not complete, just as $\mathbb{Q}$ is not complete).

Probably motivated by this, Henri Lebesgue (1875-1941) devised another theory of integration, which we now call the *Lebesgue integral*. In developing his integral, Lebesgue had to extend the notion of length to arbitrary subsets of $\mathbb{R}$. This (together with some observations of Jordan and Borel) was the birth of *measure theory* as described in the previous subsection.

As a big picture, Lebesgue integral is an extension of the Riemann integral, meaning that if a function is Riemann integrable, it is also Lebesgue integrable and its Riemann and Lebesgue integrals coincide. The class of Lebesgue integrable functions is strictly larger than the class of Riemann integrable functions.For example, the Dirichlet function has Lebesgue integral 0. In fact, Lebesgue integrable functions are the *completion* of the set of Riemann integrable functions, just like $\mathbb{R}$ is the completion of $\mathbb{Q}$.

3. Measure Theory

3.1. **Measure Spaces.**

As stated in the previous section, defining a reasonable measure on all subsets of $\mathbb{R}^n$ is not possible. We will define measures on collections of sets which have certain properties:

Definition 3.1. Let X be a set and $\mathcal{A}$ be a family of subsets of X. $\mathcal{A}$ is a *σ-algebra* (or a *σ-field*) if

(1) $\varnothing \in \mathcal{A}$,
(2) If $A \in \mathcal{A}$ then its complement $A^c = X - A \in \mathcal{A}$, (i.e., $\mathcal{A}$ is closed under complements)
(3) If (A_n) is a sequence in $\mathcal{A}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$. (i.e., $\mathcal{A}$ is closed under countable unions)

Example 3.2.

(1) For any set X, $\{\varnothing, X\}$ and the power set $\mathcal{P}(X)$ are σ-algebras.
(2) Let $X = \mathbb{N}$, then $\{\varnothing, \{0, 2, 4, 6, \ldots\}, \{1, 3, 5, \ldots\}, \mathbb{N}\}$ is a σ-algebra.
(3) Let X be an uncountable set and let $\mathcal{A} = \{Y \subseteq X \mid Y \text{ is countable or } Y^c \text{ is countable}\}$. Then $\mathcal{A}$ is a σ-algebra. [1]

Exercises 3.3. Show that a σ-algebra is closed under countable intersections, i.e., If (A_n) is a sequence in $\mathcal{A}$, then $\bigcap_{n=1}^{\infty} A_n \in \mathcal{A}$.

Exercises 3.4. Let X be a set and let $\{\mathcal{A}_i \mid i \in I\}$ be a family of σ-algebras on X. Then the intersection $\bigcap_{i \in I} \mathcal{A}_i$ is also a σ-algebra.

The previous exercise implies that one can talk about the smallest σ-algebra containing a set:

Definition 3.5. Let X be a set and let $\mathcal{Y} \subseteq \mathcal{P}(X)$. The *$\sigma$-algebra generated by $\mathcal{Y}$* (denoted by $\sigma(\mathcal{Y})$) is the intersection of all σ-algebras containing $\mathcal{Y}$. In other words,

$$\sigma(\mathcal{Y}) = \bigcap_{\substack{\mathcal{A} \text{ is a } \sigma\text{-algebra} \\ \text{containing } \mathcal{Y}}} \mathcal{A}$$

Note that, since $\mathcal{P}(X)$ is a σ-algebra containing $\mathcal{Y}$, the intersection is non-empty. Also, from the previous exercise, $\sigma(\mathcal{Y})$ is a σ-algebra. It is clearly the smallest σ-algebra containing $\mathcal{Y}$ in the sense that it is contained in any σ-algebra containing $\mathcal{Y}$.

Definition 3.6. The σ-algebra generated by the open subsets of $\mathbb{R}^n$ is the *Borel σ-algebra* of $\mathbb{R}^n$. We will denote it by $\mathcal{B}(\mathbb{R}^n)$.

We will investigate $\mathcal{B}(\mathbb{R}^n)$ in detail later.

Definition 3.7. Let X be a set and $\mathcal{A}$ a σ-algebra on X. A *measure* on X (or on $(X, \mathcal{A})$) is a function

$$\mu : \mathcal{A} \to [0, \infty]$$

satisfying

[1] Here one has to recall that a countable union of countable sets is countable.

(1) $\mu(\varnothing) = 0$
(2) for any disjoint sequence $A_1, A_2 \ldots$ of sets in $\mathcal{A}$, we have

$$\mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n) \qquad \text{(that is, } \mu \textit{ is countably additive)}$$

(Remark: the infinite sum on right of item (2) has properties as in section 1.1)

Note that countable additivity implies *finite additivity* that is, for any disjoint sets $A_1, A_2, \ldots, A_n$ we have $\mu(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} \mu(A_i)$.

Example 3.8.

(1) Given any set X and any σ-algebra $\mathcal{A}$ on X, define $\mu_1(A) = 0$ for any $A \in \mathcal{A}$. Also, define $\mu_2(A) = \begin{cases} 0 & \text{if } A = \varnothing \\ \infty & \text{if } A \neq \varnothing \end{cases}$. μ_1 and μ_2 define (totally uninteresting) measures.
(2) Let X be a set and fix a point $p \in X$. Define

$$\delta_p(A) = \begin{cases} 0 & \text{if } p \notin A \\ 1 & \text{if } p \in A \end{cases}$$

δ_p defines a measure on $\mathcal{P}(X)$ which we will call the *unit measure* (or the *Dirac measure*) concentrated at p.
(3) For a finite set X, $\mu(A) = \dfrac{|A|}{|X|}$ defines a measure on $\mathcal{P}(X)$.
(4) Let $X = \mathbb{N}$ and let $\mathcal{A} = \mathcal{P}(\mathbb{N})$. Define

$$\mu(A) = \begin{cases} n & \text{if } A \text{ has n elements} \\ \infty & \text{if } A \text{ is infinite} \end{cases}$$

Then μ is a measure on $\mathbb{N}$ which we will call the *counting measure*.

Definition 3.9. Let X be a set, $\mathcal{A}$ a σ-algebra on X and μ a measure on $\mathcal{A}$. The triple $(X, \mathcal{A}, \mu)$ is called a *measure space*.

Here are some finiteness properties of measure spaces.

Definition 3.10. Let $(X, \mathcal{A}, \mu)$ be a measure space.

(1) μ is called *finite* if $\mu(X) < \infty$.
(2) μ is called *σ-finite* if there is a sequence $E_1, E_2, \ldots$ in $\mathcal{A}$ such that $\mu(E_n) < \infty$ for any n and $X = \bigcup_{n=1}^{\infty} E_n$.

Clearly a finite measure is σ-finite. The counting measure on $\mathbb{N}$ is σ-finite but not finite.

Here are some properties which are true in a general measure space:

Lemma 3.11. *Let $(X, \mathcal{A}, \mu)$ be a measure space, $E, F \in \mathcal{A}$ with $E \subseteq F$ and (E_n) a sequence in $\mathcal{A}$, then*

(1) *$\mu(E) \leq \mu(F)$. (i.e., μ is monotone)*
(2) *if $\mu(E) < \infty$, then $\mu(F - E) = \mu(F) - \mu(E)$*[1]

[1]Here we have the convention that $\infty - x = \infty$ for any $x \in \mathbb{R}$.

(3) $\mu(\bigcup_{n=1}^{\infty} E_n) \leq \sum_{n=1}^{\infty} \mu(E_n)$ *(i.e., μ is countably sub-additive)*

Proof. Since $F = E \cup (F - E)$ and $E \cap (F - E) = \varnothing$, $\mu(F) = \mu(E) + \mu(F - E) \geq \mu(E)$. Also, if $\mu(E) < \infty$, then (subtracting) we get $\mu(F - E) = \mu(F) - \mu(E)$. For (3), let $F_1 = E_1$ and $F_n = E_n - \bigcup_{k=1}^{n-1} E_n$ for $n > 1$. Then, (F_n) is a disjoint sequence with $\bigcup_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} F_n$. So,

$$\mu(\bigcup_{n=1}^{\infty} E_n) = \mu(\bigcup_{n=1}^{\infty} F_n) = \sum_{n=1}^{\infty} \mu(F_n) \leq \sum_{n=1}^{\infty} \mu(E_n)$$

□

Definition 3.12. Let $A_1, A_2, \ldots$ be a sequence of subsets of a set. The sequence is *increasing* if $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots A_n \subseteq A_{n+1} \cdots$ and *decreasing* if $A_1 \supseteq A_2 \supseteq A_3 \cdots \supseteq A_n \supseteq A_{n+1} \cdots$

Lemma 3.13. *Let $(X, \mathcal{A}, \mu)$ be a measure space.*

(1) *If (E_n) is an increasing sequence, then* $\mu(\bigcup_{n=1}^{\infty} E_n) = \lim_{n\to\infty} \mu(E_n)$

(2) *If (F_n) is a decreasing sequence with $\mu(F_1) < \infty$, then* $\mu(\bigcap_{n=1}^{\infty} F_n) = \lim_{n\to\infty} \mu(F_n)$

Proof. (1) Note that if $\mu(E_k) = \infty$ for some k, then $\mu(E_k) = \infty$ for all $n \geq k$, thus both sides are of the asserted equality are ∞. So, assume that $\mu(E_n) < \infty$ for all n. Let $A_1 = E_1$ and $A_n = E_n - E_{n-1}$ for $n > 1$. Note that (A_n) is disjoint, for each $n \geq 1$ we have $E_n = \bigcup_{i=1}^{n} A_i$ and $\bigcup_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} A_n$. So,

$$\mu(\bigcup_{n=1}^{\infty} E_n) = \mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n) = \lim_{N\to\infty} \sum_{n=1}^{N} \mu(A_n) = \lim_{N\to\infty} \mu(\bigcup_{n=1}^{N} A_n) = \lim_{N\to\infty} \mu(E_N)$$

(2) Note that the sequence $(F_1 - F_n)$ is an increasing sequence. So, using Lemma 3.11 and part (1) of this Lemma, we have

$$\mu(F_1) - \mu(\bigcap_{n=1}^{\infty} F_n) = \mu(F_1 - \bigcap_{n=1}^{\infty} F_n) = \mu\left(\bigcup_{n=1}^{\infty} (F_1 - F_n)\right) = \lim_{n\to\infty} \mu(F_1 - F_n) = \mu(F_1) - \lim_{n\to\infty} \mu(F_n)$$

and hence $\mu(\bigcap_{n=1}^{\infty} F_n) = \lim_{n\to\infty} \mu(F_n)$. □

For the counting measure μ on $\mathbb{N}$, let $F_n = \{n, n+1, \ldots\}$. We have $\bigcap_{n=1}^{\infty} F_n = \varnothing$ but $\mu(F_n) = \infty$ for all n. Thus, item (2) may fail if $\mu(F_1) = \infty$.

3.1.1. *Exercises.*

(1) Show that a σ-algebra is closed under countable intersections, i.e., If (A_n) is a sequence in $\mathcal{A}$, then $\bigcap_{n=1}^{\infty} A_n \in \mathcal{A}$.

(2) Let X be a set and let $\{\mathcal{A}_i \mid i \in I\}$ be a family of σ-algebras on X. Then the intersection $\bigcap_{i\in I} \mathcal{A}_i$ is also a σ-algebra.
(3) Let X be an uncountable set and let $\mathcal{A} = \{Y \subseteq X \mid Y \text{ is countable or } Y^c \text{ is countable}\}$. Show that $\mathcal{A}$ is a σ-algebra.
(4) Show that $[0, 1) \in \mathcal{B}(\mathbb{R})$.
(5) Show that $\mathcal{B}(\mathbb{R}^n)$ is the σ-algebra generated by open balls. (Hint: You can use Lemma 3.31 below)
(6) Show that $\mathcal{B}(\mathbb{R}^n)$ is also the σ-algebra generated by the closed subsets of $\mathbb{R}^n$.
(7) (This is the "disjointification trick" that we are using frequently)
Let X be a set and (A_k) be a sequence of subsets of X. Define $E_1 = A_1$ and for $n > 1$, $E_k = A_k - \bigcup_{i=1}^{k-1} A_i$. Show that (E_k) is a sequence of disjoint sets and $\bigcup_{k=1}^{\infty} A_k = \bigcup_{k=1}^{\infty} E_k$.
(8) Convince yourself that the functions in Example 3.8 are really measures.
(9) Let $(X, \mathcal{A}, \mu)$ be a measure space. Fix $E \in \mathcal{A}$ and define $\mu_E(A) = \mu(E \cap A)$ for any $A \in \mathcal{A}$. Show that μ_E is a measure on $\mathcal{A}$.
(10) Let $X = \mathbb{N}$ and $\mathcal{A} = \mathcal{P}(\mathbb{N})$. Fix a sequence (a_n) of non-negative real numbers. Define $\mu : \mathcal{P}(\mathbb{N}) \to [0, \infty]$ by

$$\mu(\varnothing) = 0 \ \ , \mu(E) = \sum_{n\in E} a_n \text{ if } E \neq \varnothing$$

Show that μ is a measure.

3.2. **Volumes of Cells and Intervals.**

Having seen some general properties of measures, we now have a specific goal as stated on various occasions: We want to extend the notions of length, area, volume etc. to larger classes of subsets. Firstly, we need to talk about sets which are the building blocks. The material of this subsection will be used in the next subsection.

Definition 3.14.

(1) Let $a, b \in \mathbb{R}$ with $a \leq b$. Each of the following types subsets of $\mathbb{R}$ will be called a *cell* in $\mathbb{R}$ (with endpoints a and b):

$$(a, b) \ , \ [a, b] \ , \ (a, b] \ , \ [a, b)$$

(2) Let A be a cell with end points a, b. Its *length* is defined as $\ell(A) = b - a$.

(If $a = b$ we have $(a, b) = (a, b] = [a, b) = \varnothing$ and $[a, b] = \{a\}$.)

As usual, cells of the first type are called *open*, of the second type are called *closed* and the remaining are *half-open* (or half-closed). This makes sense since cells of type 1 are open subsets of $\mathbb{R}$ and cells of type 2 are closed subsets of $\mathbb{R}$. The remaining are neither open nor closed subsets.

The collection of all cells of type $(a, b]$ have the following nice property: The intersection,union and difference two such sets are either of this type or the *disjoint union* of such sets. Also, the complement of such a set is a *countable union* of sets of the same type, i.e.,

$$\mathbb{R} - (a,b] = (-\infty, a] \cup (b, \infty) = \bigcup_{n=1}^{\infty} \left(a - n, a - \frac{1}{n} \right] \ \cup \ \bigcup_{n=1}^{\infty} \left(b - \frac{1}{n}, b + n \right]$$

(The same is true for the collection of sets of the form $[a,b)$)

Note that cells are *intervals* in $\mathbb{R}$. But we do not allow cells to have end points $\pm\infty$. If this happens we have the infinite intervals :

$$(-\infty, a) \ , \ (-\infty, a] \ , \ [b, \infty) \ , \ (b, \infty) \ , \ (-\infty, \infty)$$

We define the length of an infinite interval to be ∞.

Now, let us generalize these to $\mathbb{R}^n$ for $n \geq 1$.

Definition 3.15.

(1) Let $n > 1$. A *cell* in $\mathbb{R}^n$ is the Cartesian product of n cells in $\mathbb{R}$. i.e., a set of the form
$$I = I_1 \times I_2 \times \cdots \times I_n$$
where $I_1, I_2, \ldots, I_n$ are cells in $\mathbb{R}$.

(2) If $I = I_1 \times I_2 \times \cdots I_n$ is a cell in $\mathbb{R}^n$, we define the *n-dimensional volume* of I to be
$$\ell(I) = \ell(I_1) \cdot \ell(I_2) \cdots \ell(I_n)$$
i.e., if I_j has end points $a_j \leq b_j$, then $\ell(I) = (b_1 - a_1)(b_2 - a_2) \cdots (b_n - a_n)$

The collection of all cells which are products of type $(a,b]$ have the following nice property: The intersection,union and difference two such sets are either of this type or the *disjoint union* of such sets. Also, the complement of such a set is a *countable union* of sets of the same type. (The same is true for the collection of cells which are products of the form $[a,b)$)

We define an *interval* in $\mathbb{R}^n$ to be a set of the form $I = I_1 \times I_2 \times \cdots I_n$ where each I_j is an interval in $\mathbb{R}$. The n-dimensional volume of such a set is $\ell(I) = \ell(I_1) \cdot \ell(I_2) \cdots \ell(I_n)$. Here one must take caution: We have $\ell(I) = 0$ if at least one of the $\ell(I_j) = 0$ and $\ell(I) = \infty$ if all $\ell(I_j) \neq 0$ and at least one $\ell(I_k) = \infty$.

So, for every $n \geq 1$ and every interval $I \subseteq \mathbb{R}^n$ we have defined its n-dimensional volume $\ell(I) \in [0, \infty]$.

For any cell $I \in \mathbb{R}^n$, let I° be its *interior* and let $\bar{I}$ be its *closure*. For example, if $I = (a,b]$, then $I^\circ = (a,b)$ and $\bar{I} = [a,b]$. If $I = [a,b) \times (c,d) \in \mathbb{R}^2$, then $I^\circ = (a,b) \times (c,d)$ and $\bar{I} = [a,b] \times [c,d]$.

Note that for any cell I, we have $\ell(I) = \ell(I^\circ) = \ell(\bar{I})$.

3.3. **The Outer Measure.**

Now that we can "measure" cells and intervals, we aim to extend it to more complicated sets. Clearly if a set is a disjoint union of finitely many cells, its measure should be the sum of the measures of the cells involved. But, there is a caution here: One needs to show this does not depend on the particular decomposition of the set into cells. Also, not every subset is a finite union of cells (for examples the unit disk in $\mathbb{R}^2$). Thus, one needs to consider countable unions. Once again one needs to show that this does not depend on the decomposition.

The next notion is similar to the upper Riemann sums in estimating the area under a curve.

Definition 3.16. Let $E \subseteq \mathbb{R}^n$. The *(Lebesgue) outer measure* of E is $m^*(E) \in [0, \infty]$ defined by

$$m^*(E) = \inf \left\{ \sum_{k=1}^{\infty} \ell(I_k) \mid (I_k) \text{ is a sequence of cells in } \mathbb{R}^n \text{ with } E \subseteq \bigcup_{k=1}^{\infty} I_k \right\}$$

(In words, the set E is covered by a countable family of cells so that the total volume of cells provides an approximation for the "volume of E". Then we take the infimum over all such coverings.)

Remark 3.17.

(1) Since $\mathbb{R}^n$ can be covered by a countable family of cells, the set over which we take infimum is non-empty. Thus m^* is well defined and $m^*(E) \geq 0$.
(2) Note that $m^*(\mathbb{R}^n) = \infty$. (This follows directly from properties of m^* in Theorem 3.18 below.)
(3) Since the terms of the summation $\sum_{k=1}^{\infty} \ell(I_k)$ are non-negative, this series converges absolutely or is equal to ∞. So, the order of summation does not change its value. Hence instead of "sequence" we could write "countable family" in the definition of m^*.
(4) In the definition of m^*, we could replace "cell" with "closed cell", i.e., we could define m^* as

$$m^*(E) = \inf \left\{ \sum_{k=1}^{\infty} \ell(I_k) \mid (I_k) \text{ is a sequence of closed cells in } \mathbb{R}^n \text{ with } E \subseteq \bigcup_{k=1}^{\infty} I_k \right\}$$

This would not change the definition since $I \subseteq \bar{I}$ and $\ell(I) = \ell(\bar{I})$ for any cell I.

(Here is a complete argument: Let $a = M^*(E)$ as in the original definition and $b = m^*(E)$ defined using closed cells. Since each closed cell is also a cell, we have $a \leq b$. For $\epsilon > 0$, there is a sequence of cells (I_k) with $E \subseteq \bigcup_{k=1}^{\infty} I_k$ and $\sum_{k=1}^{\infty} \ell(I_k) \leq a + \epsilon$. Since $\bigcup_{k=1}^{\infty} I_k \subseteq \bigcup_{k=1}^{\infty} \bar{I}_k$, we have $b \leq \sum_{k=1}^{\infty} \ell(\bar{I}_k) = \sum_{k=1}^{\infty} \ell(I_k) \leq a + \epsilon$. Since ϵ is arbitrary, we have $b \leq a$.)

(5) In the definition of m^*, we could replace "cell" with "open cell" (or with "half-open cell")

(Here is a complete argument: Let $a = m^*(E)$ as in the original definition and $b = m^*(E)$ defined using open (or half-open) cells. Since each open (or half-open) cell is also a cell, we have $a \leq b$. For $\epsilon > 0$, there is a sequence of cells (I_k) with $E \subseteq \bigcup_{k=1}^{\infty} I_k$ and $\sum_{k=1}^{\infty} \ell(I_k) \leq a + \frac{\epsilon}{2}$. Let (J_k) be a sequence of open (or half-open) cells such that $I_k \subseteq J_k$ and $\ell(J_k) \leq \ell(I_k) + \frac{\epsilon}{2^{k+1}}$ for any $k \geq 1$. So, we have $b \leq \sum_{k=1}^{\infty} \ell(J_k) \leq \sum_{k=1}^{\infty} \ell(I_k) + \frac{\epsilon}{2} \leq a + \epsilon$. Since ϵ is arbitrary, we have $b \leq a$.)

Recall that if $A \subseteq \mathbb{R}^n$, the *diameter* of A is $diam(A) = \sup\{\|x - y\| \mid x, y \in A\}$

(6) For $\delta > 0$, in the definition of m^* one can replace "cells" with "cells of diameter $< \delta$"

(Hint: If $(I_k)$is a sequence of cells covering E, each cell I_k is the union of finitely many cells of diameter $< \delta$. So, there is a sequence of cells (J_k) of diameter $< \delta$ such that $\bigcup I_k = \bigcup J_k$.)

Theorem 3.18. *The function $m^* : \mathcal{P}(\mathbb{R}^n) \to [0, \infty]$ satisfies*

(1) *If $E \subseteq F$ then $m^*(E) \leq m^*(F)$ (monotonicty)*
(2) *If (E_k) is a sequence of subsets of $\mathbb{R}^n$, then*

$$m^*(\bigcup_{k=1}^{\infty} E_k) \leq \sum_{k=1}^{\infty} m^*(E_k) \qquad \textit{(i.e., } m^* \textit{ is countably sub-additive)}$$

Proof. (1) If $E \subseteq F$, any covering of F by unions of cells gives a covering of E by cells, thus $m^*(E) \leq m^*(F)$.

(2) If for some k we have $m^*(E_k) = \infty$ then the assertion is trivially true. So assume $m^*(E_k) < \infty$ for all k.

Fix $\epsilon > 0$. For each $k \geq 1$, let (I_j^k) be a covering of E_k by cells such that $\sum_{j=1}^{\infty} \ell(I_j^k) \leq m^*(E_k) + \frac{\epsilon}{2^k}$. It follows that $\{I_j^k \mid k \geq 1, j \geq 1\}$ is a covering of $\bigcup_{k=1}^{\infty} E_k$ by cells. Hence, (using Theorem 1.2)

$$m^*(\bigcup_{k=1}^{\infty} E_k) \leq \sum_{k\geq 1, j\geq 1} \ell(I_j^k) = \sum_{k=1}^{\infty}\sum_{j=1}^{\infty} \ell(I_j^k) \leq \sum_{k=1}^{\infty} \left(m^*(E_k) + \frac{\epsilon}{2^k}\right) = \sum_{k=1}^{\infty} m^*(E_k) + \epsilon$$

The result follows since ϵ is arbitrary. □

It will be observed at the end of this section that m^* cannot be countably additive or finitely additive. But for sets which have *positive distance*, it is additive.

If $A, B \subseteq \mathbb{R}^n$, define the *distance* between A and B as

$$\operatorname{dist}(A, B) = \inf\{\|a - b\| \mid a \in A, b \in B\}$$

Theorem 3.19. *Let $A, B \subseteq \mathbb{R}^n$ with $\operatorname{dist}(A, B) > 0$. Then $m^*(A \cup B) = m^*(A) + m^*(B)$.*

Proof. The fact that $m^*(A \cup B) \leq m^*(A) + m^*(B)$ follows from the previous theorem. Thus, we may assume $m^*(A \cup B) < \infty$.

For the converse inequality, let $\operatorname{dist}(A, B) = \delta > 0$ and fix $\epsilon > 0$. By virtue of Remark 3.17 (5), let (I_k) be a covering of $A \cup B$ by cells of diameter $< \delta$ such that $\sum_{k=1}^{\infty} \ell(I_k) \leq m^*(A \cup B) + \epsilon$. It follows that if for some $k, A \cap I_k \neq \varnothing$, then $B \cap I_k = \varnothing$ and vice versa. Define $X = \{k \mid I_k \cap A \neq \varnothing\}$ and $Y = \{k \mid I_k \cap B \neq \varnothing\}$ so that $X \cap Y = \varnothing$. Hence $\{I_k \mid k \in X\}$ is a covering of A and $\{I_k \mid k \in Y\}$ is a covering of B by cells. Thus,

$$m^*(A) + m^*(B) \leq \sum_{k\in X} \ell(I_k) + \sum_{k\in Y} \ell(I_k) \leq \sum_{k=1}^{\infty} \ell(I_k) \leq m^*(A \cup B) + \epsilon$$

Surprisingly, this proof also ends with the phrase "The result follows since ϵ is arbitrary" :) □

The next theorem shows that for a cell, its outer measure and its volume are equal. For this we need the following lemma:

Lemma 3.20. *Let $I \subseteq \mathbb{R}^n$ be a cell which is covered by the cells $I_1, \ldots, I_r$. Then, $\ell(I) \leq \sum_{k=1}^{r} \ell(I_k)$.*

Proof. Suppose the cells $I_1, \ldots, I_r \subseteq \mathbb{R}^n$ cover the cell $I \subseteq \mathbb{R}^n$.

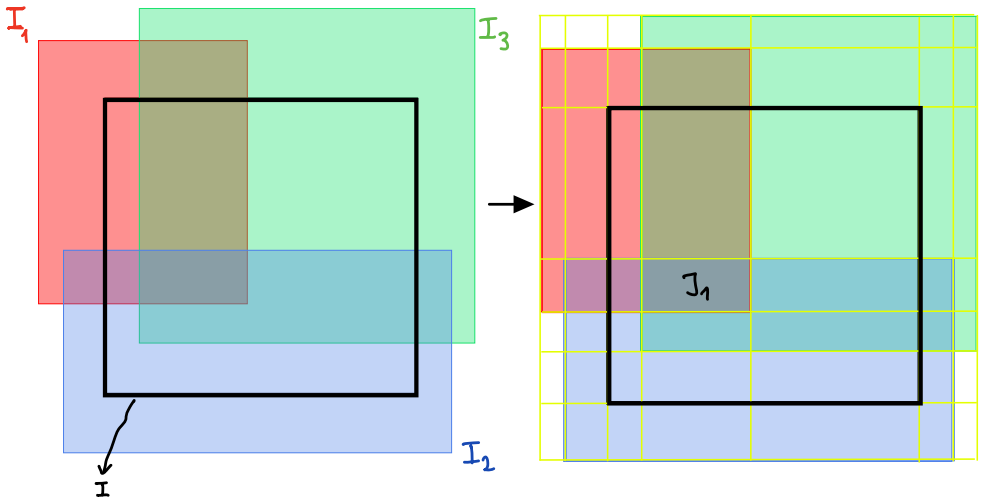
For a fixed coordinate $i \in \{1, \ldots, n\}$, the end points of the i-th *side* of each of the cells $I, I_1, I_2, \ldots, I_r$ partition $\mathbb{R}$ into intervals. Doing this for each coordinate $i \in \{1, 2, \ldots, n\}$ and then considering the (smaller) cells whose sides correspond to the finite intervals, one obtains a family of cells $J_1, J_2, \ldots, J_s$ and sets A_k with $\{1, \ldots, s\} = A_1 \cup A_2 \cup \cdots \cup A_r$ such that

$$I \subseteq \bigcup_{t=1}^{r} J_t \quad \text{and} \quad I_k = \bigcup_{t \in A_k} J_t \quad \text{for every} \quad k = 1, 2, \ldots, r$$

(See the picture below). Moreover, the cells $J_1, J_2, \ldots, J_s$ have disjoint interiors. Because of this, for each k we have $\ell(I_k) = \sum_{t \in A_k} \ell(J_t)$ and if $A = \{t \mid J_t^\circ \cap I^\circ \neq \varnothing\}$, then $\ell(I) = \sum_{t \in A} \ell(J_t)$. Hence

$$\sum_{k=1}^{r} \ell(I_k) = \sum_{k=1}^{r} \sum_{t \in A_k} \ell(J_t) \geq \sum_{t \in A} \ell(J_t) = \ell(I).$$

□



Theorem 3.21. *If $I \subseteq \mathbb{R}^n$ is a cell, then $m^*(I) = \ell(I)$.*

Proof. Since I itself is a covering of I, we have $m^*(I) \leq \ell(I)$.

Conversely, for $\epsilon > 0$, let (I_k) be a covering of I by open cells such that $\sum_{k=1}^{\infty} \ell(I_k) \leq m^*(I) + \epsilon/2$. Let J be a closed cell such that $J \subseteq I$ and $\ell(I) - \epsilon/2 < \ell(J)$. Since J is compact, finitely many

$I_1, \ldots, I_r$ cover J [1]. Now apply the previous lemma to J, to get

$$\ell(J) \leq \sum_{k=1}^{r} \ell(I_k) \leq \sum_{k=1}^{\infty} \ell(I_k) \leq m^*(I) + \epsilon/2$$

and hence $\ell(I) \leq \ell(J) + \epsilon/2 \leq m^*(I) + \epsilon$. □

Given a subset $E \subseteq \mathbb{R}^n$ and $x \in \mathbb{R}^n$, define the *translate* of E by x as $x + E = \{x + e \mid e \in E\}$. The next theorem shows that the outer measure m^* is *translation invariant*:

Theorem 3.22. *For any $E \subseteq \mathbb{R}^n$ and $x \in \mathbb{R}^n$, $m^*(x + E) = m^*(E)$.*

Proof. Note that if I is a cell, then its translate $x + I$ is also a cell with $\ell(I) = \ell(x + I)$. Thus if (I_k) is a covering of E by cells, then $(x + I_k)$ is a covering of $x + E$ by cells. So, for $\epsilon > 0$, let (I_k) be a covering of E with $\sum_{k=1}^{\infty} \ell(I_k) \leq m^*(E) + \epsilon$.

Then $m^*(x + E) \leq \sum_{k=1}^{\infty} \ell(x + I_k) = \sum_{k=1}^{\infty} \ell(I_k) \leq m^*(E) + \epsilon$ and hence $m^*(x + E) \leq m^*(E)$ The reverse inequality follows by symmetry since, E is the translate of $x + E$ by $-x$. □

Remark 3.23. It is clear that the outer measure is not countably additive (See section 2.2). Here is an argument that shows it is also not finitely additive:

Recall the *Vitali set* $V \subseteq [0, 1]$ from section 2.2 and its translates let $V_i = V + q_i$ where $\mathbb{Q} \cap [-1, 1] = \{q_1, q_2, \ldots\}$. Recall also that $[0, 1] \subseteq E = \bigcup_{i=1}^{\infty} V_i \subseteq [-1, 2]$ and also $V_i \subseteq [-1, 2]$ if $i \neq j$ then $V_i \cap V_j = \varnothing$.

Note that, $1 \leq m^*(E) \leq \sum_{i=1}^{\infty} m^*(V_i) = \sum_{i=1}^{\infty} m^*(V)$ and hence $m^*(V) > 0$.

Let $n > 0$ large enough so that $1/n < m^*(V)$.

Suppose now that m^* is finitely additive.

Let $I \subseteq \mathbb{Q} \cap [-1, 1]$ with $|I| = 3n$. Then

$$m^*\left(\bigcup_{i \in I} V_i\right) = \sum_{i \in I} m^*(V_i) = \sum_{i \in I} m^*(V) > 3n \cdot 1/n = 3.$$

But $m^*\left(\bigcup_{i \in I} V_i\right) \leq m^*(E) \leq 3$. Hence we get a contradiction.

Exercises 3.24.

(1) Write a complete proof for the last item of Remark 3.17.

3.4. **Some words on the Jordan Outer Measure.**

We could define the outer measure by replacing countable unions of cells by finite unions of cells to get:

[1] We are using the Heine-Borel Theorem, See subsection 1.2

Definition 3.25. Let $E \subseteq \mathbb{R}^n$ be bounded. The *Jordan outer measure* (or the *Jordan content*) of E is $J^*(E) \in [0, \infty)$ defined by

$$J^*(E) = \inf \left\{ \sum_{k=1}^{N} \ell(I_k) \mid I_1, I_2, \ldots, I_N \text{ are cells in } \mathbb{R}^n \text{ with } E \subseteq \bigcup_{k=1}^{N} I_k \right\}$$

Note that m^* and J^* are are in general different, for example, let us show $m^*(\mathbb{Q} \cap [0,1]) = 0$ but $J^*(\mathbb{Q} \cap [0,1]) = 1$:

Let $\mathbb{Q} \cap [0,1] = \{q_1, q_2, \ldots\}$. For $\epsilon > 0$ pick an interval I_k of length $\frac{\epsilon}{2^k}$ around each q_k. So $m^*(\mathbb{Q} \cap [0,1]) \leq \sum_{k=1}^{\infty} \ell(I_k) = \sum_{k=1}^{\infty} \frac{\epsilon}{2^k} = \epsilon$. Thus, $m^*(\mathbb{Q} \cap [0,1]) = 0$.

One can show (by an argument which is similar to that of m^*) that J^* is monotone. Therefore $J^*(\mathbb{Q} \cap [0,1]) \leq J^*([0,1]) = 1$. Suppose that $J^*(\mathbb{Q} \cap [0,1]) < 1$. This means there are cells $I_1, \ldots, I_N$ such that $\sum_{k=1}^{N} \ell(I_k) < 1$ and $\mathbb{Q} \cap [0,1] \subseteq \bigcup_{k=1}^{N} I_k$. Note that we can't have $[0,1] \subseteq \bigcup_{k=1}^{N} I_k$ since then we would have $\sum_{k=1}^{N} \ell(I_k) \geq 1$. Then (since $\mathbb{Q} \cap [0,1]$ is dense in $[0,1]$) there must be a rational number in $(\mathbb{Q} \cap [0,1]) - \bigcup_{k=1}^{N} I_k$. This is a contradiction.

It follows that the outer measure J^* is not countably sub-additive. (The set $\mathbb{Q} \cap [0,1]$ is countable union of sets with one element.)

This measure is related to the Riemann Integral (more precisely to upper Riemann Sums) as we shall discuss in the later sections.

See [3] for properties of the Jordan outer measure.

3.5. **Measurable Sets.**

As noted in the earlier sections, the aim is to determine sets which are *measurable*. There are several equivalent definitions. We will use the following criterion due to C. Carathéodory, let us call it the *Carathéodory condition*:

Definition 3.26. Let m^* be the outer measure on $\mathbb{R}^n$ as defined in Section 3.3. A subset $E \subseteq \mathbb{R}^n$ is (Lebesgue) *measureable* if

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c) \text{ for all subsets } A \subseteq \mathbb{R}^n$$

This needs to be explained as it is not immediately clear why such definition is useful. (The original development of Lebesgue uses the notion of an *inner measure* and defines a set to be measurable if the inner and outer measures of the set coincide. This is similar to the lower/upper sums of Riemann integration. As most books on this subject, we will not follow this route.)

Recall that our aim is to find a criterion for a set to be not pathological in the measure sense. In the Carathéodory condition, if one picks $A = \mathbb{R}^n$, the equation becomes $m^*(\mathbb{R}^n) = m^*(E) + m^*(E^c)$ which is a reasonable expectation from a "nice" set. The Carathéodory condition requires that this equality holds not just for $\mathbb{R}^n$, but for every subset of $\mathbb{R}^n$. In other words, the set E is required to split *every* subset A "nicely" so that we have $m^*(A) = m^*(A \cap E) + m^*(A \cap E^c)$.

One can simplify the Carathéodory condition:

Lemma 3.27. *$E \subset \mathbb{R}^n$ is measurable if and only if for any $A \subseteq \mathbb{R}^n$ with $m^*(A) < \infty$, we have $m^*(A) \geq m^*(A \cap E) + m^*(A \cap E^c)$*

Proof. Note that $A = (A \cap E) \sqcup (A \cap E^c)$ and hence $m^*(A) \leq m^*(A \cap E) + m^*(A \cap E^c)$ holds always. So if $m^*(A) = \infty$, the Carathéodory condition holds trivially. □

Theorem 3.28. *Let $\mathcal{L}(\mathbb{R}^n) \subseteq \mathcal{P}(\mathbb{R}^n)$ be the collection of measurable subsets of $\mathbb{R}^n$. (i.e., $\mathcal{L}(\mathbb{R}^n)$ is the collection of subsets which satisfy the Carathéodory condition). Then,*

1. *$\mathcal{L}(\mathbb{R}^n)$ is a σ-algebra,*
2. *The restriction of m^* to $\mathcal{L}(\mathbb{R}^n)$ is a measure.*

Proof. It is clear that $\varnothing \in \mathcal{L}(\mathbb{R}^n)$ and it is also clear that $\mathcal{L}(\mathbb{R}^n)$ is closed under complements. It remains to show that $\mathcal{L}(\mathbb{R}^n)$ is closed under countable unions. We will achieve this in several steps.

Let us first show that if $E, F \in \mathcal{L}(\mathbb{R}^n)$ then $E \cap F \in \mathcal{L}(\mathbb{R}^n)$.

Let $A \subseteq \mathbb{R}^n$, since $E \in \mathcal{L}(\mathbb{R}^n)$ we have:

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c)$$

and since $F \in \mathcal{L}(\mathbb{R}^n)$ we have

$$m^*(A \cap E) = m^*(A \cap E \cap F) + m^*(A \cap E \cap F^c)$$

Combining these:

$$m^*(A) = m^*(A \cap E \cap F) + m^*(A \cap E \cap F^c) + m^*(A \cap E^c) \tag{1}$$

Also, since $E \in \mathcal{L}(\mathbb{R}^n)$,

$$m^*(A \cap (E \cap F)^c) = m^*(A \cap (E \cap F)^c \cap E) + m^*(A \cap (E \cap F)^c \cap E^c)$$

$$= m^*(A \cap F^c \cap E) + m^*(A \cap E^c) \tag{2}$$

Combining (1) and (2) we get, $m^*(A) = m^*(A \cap (E \cap F)) + m^*(A \cap (E \cap F)^c)$ for any $A \subseteq \mathbb{R}^n$, hence $E \cap F \in \mathcal{L}(\mathbb{R}^n)$.

Since $\mathcal{L}(\mathbb{R}^n)$ is closed under complements and intersections, it follows that if $E, F \in \mathcal{L}(\mathbb{R}^n)$ then $E \cup F = (E \cap F)^c \in \mathcal{L}(\mathbb{R}^n)$, that is, $\mathcal{L}(\mathbb{R}^n)$ is also closed under unions. A simple induction now shows that if $E_1, E_2, \ldots, E_k \in \mathcal{L}(\mathbb{R}^n)$ then $E_1 \cup \cdots \cup E_k \in \mathcal{L}(\mathbb{R}^n)$.

Claim1: If $E_1, E_2, \ldots, E_k \in \mathcal{L}(\mathbb{R}^n)$ are *disjoint*, then for all $A \subseteq \mathbb{R}^n$,

$$m^*(A \cap \bigcup_{i=1}^{k} E_i) = \sum_{i=1}^{k} m^*(A \cap E_i)$$

Arguing inductively, for $k = 1$ this is clear. For $k > 1$, using the fact that $E_1 \in \mathcal{L}(\mathbb{R}^n)$, we have

$$m^*(A \cap \bigcup_{i=1}^{k} E_i) = m^*(A \cap \bigcup_{i=1}^{k} E_i \cap E_1) + m^*(A \cap \bigcup_{i=1}^{k} E_i \cap E_1^c) = m^*(A \cap E_1) + m^*(A \cap \bigcup_{i=2}^{k} E_i)$$

(where in the last equality we use the fact that $E_1, E_2, \ldots, E_k$ disjoint). This proves the claim.

Claim 2: If (E_i) is a sequence of *disjoint* sets in $\mathcal{L}(\mathbb{R}^n)$, then $\bigcup_{i=1}^{\infty} E_i \in \mathcal{L}(\mathbb{R}^n)$.

For $A \subseteq \mathbb{R}^n$, since $\bigcup_{i=1}^{k} E_i \in \mathcal{L}(\mathbb{R}^n)$ we have (using claim 1)

$$m^*(A) = m^*(A \cap \bigcup_{i=1}^{k} E_i) + m^*(A \cap (\bigcup_{i=1}^{k} E_i)^c) = \sum_{i=1}^{k} m^*(A \cap E_i) + m^*(A \cap (\bigcup_{i=1}^{k} E_i)^c)$$

Since $\bigcup_{i=1}^{k} E_i \subseteq \bigcup_{i=1}^{\infty} E_i$ we have $A \cap (\bigcup_{i=1}^{\infty} E_i)^c \subseteq A \cap (\bigcup_{i=1}^{k} E_i)^c$. Thus:

$$m^*(A) \geq \sum_{i=1}^{k} m^*(A \cap E_i) + m^*(A \cap (\bigcup_{i=1}^{\infty} E_i)^c)$$

This inequality holds for every k, thus taking limit as $k \to \infty$ we have

$$m^*(A) \geq \sum_{i=1}^{\infty} m^*(A \cap E_i) + m^*\left(A \cap (\bigcup_{i=1}^{\infty} E_i)^c\right) \text{ for any } A \subseteq \mathbb{R}^n \tag{3}$$

Combining this with the fact that

$$m^*(A \cap \bigcup_{i=1}^{\infty} E_i) = m^*(\bigcup_{i=1}^{\infty} A \cap E_i) \leq \sum_{i=1}^{\infty} m^*(A \cap E_i) \quad (m^* \text{ is sub-additive!})$$

we get

$$m^*(A) \geq m^*(A \cap \bigcup_{i=1}^{\infty} E_i) + m^*(A \cap (\bigcup_{i=1}^{\infty} E_i)^c) \text{ for any } A \subseteq \mathbb{R}^n$$

Thus, from the previous lemma we have that $\bigcup_{i=1}^{\infty} E_i \in \mathcal{L}(\mathbb{R}^n)$.

In words we have shown that $\mathcal{L}(\mathbb{R}^n)$ is closed under countable *disjoint* unions.

To see that $\mathcal{L}(\mathbb{R}^n)$ is closed under countable unions we use a familiar argument:

Let (E_i) be a sequence in $\mathcal{L}(\mathbb{R}^n)$. Letting $F_1 = E_1$ and $F_i = E_i - \bigcup_{j=1}^{i-1} E_j = E_i \cap \left(\bigcup_{j=1}^{i-1} E_j\right)^c$, we see that (F_i) is a disjoint sequence in $\mathcal{L}(\mathbb{R}^n)$ with $\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} F_i$. Thus, $\bigcup_{i=1}^{\infty} E_i \in \mathcal{L}(\mathbb{R}^n)$. This proves part 1 of the theorem.

Next we will show that, when restricted to $\mathcal{L}(\mathbb{R}^n)$, m^* defines a measure. Clearly $m^*(\varnothing) = 0$, thus we need to show that m^* is countably additive on $\mathcal{L}(\mathbb{R}^n)$. Let (E_i) be a disjoint sequence in $\mathcal{L}(\mathbb{R}^n)$. Applying equation (3) with $A = \bigcup_{j=1}^{\infty} E_j$ we get

$$m^*(\bigcup_{j=1}^{\infty} E_j) \geq \sum_{i=1}^{\infty} m^*\left(\bigcup_{j=1}^{\infty} E_j \cap E_i\right) + m^*\left(\bigcup_{j=1}^{\infty} E_j \cap (\bigcup_{i=1}^{\infty} E_i)^c\right) = \sum_{i=1}^{\infty} m^*(E_i)$$

Thus, by sub-additivity of m^*, we have $m^*(\bigcup_{j=1}^{\infty} E_j) = \sum_{i=1}^{\infty} m^*(E_i)$. □

Let us denote the restriction of m^* to $\mathcal{L}(\mathbb{R}^n)$ by m, so that m is a measure on $\mathcal{L}(\mathbb{R}^n)$ by the previous theorem. In other words, if $E \in \mathcal{L}(\mathbb{R}^n)$, then $m(E) = m^*(E)$.

Definition 3.29.

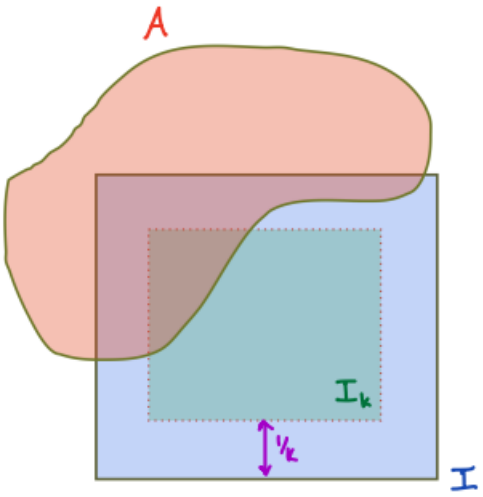
(1) The σ-algebra $\mathcal{L}(\mathbb{R}^n)$ is the *Lebesgue σ-algebra* on $\mathbb{R}^n$.
(2) Sets in $\mathcal{L}(\mathbb{R}^n)$ are the *(Lebesgue) measurable subsets* of $\mathbb{R}^n$.
(3) The measure m is the *Lebesgue Measure* on $\mathbb{R}^n$.

The next theorem shows that cells in $\mathbb{R}^n$ are measurable.

Theorem 3.30. *If $I \subseteq \mathbb{R}^n$ is a cell, then I is measurable and hence $m(I) = \ell(I)$.*

Proof. Using the condition measurability of Lemma 3.27, let $A \subseteq \mathbb{R}^n$ with $m^*(A) < \infty$. We need show that $m^*(A) \geq m^*(A \cap I) + m^*(A \cap I^c)$. (Recall that m^* is monotone and sub-additive by Theorem 3.18).

For each $k \geq 1$, let $I_k = \{x \in I \mid \text{dist}(x, I^c) > 1/k\}$ (see the picture below).



For each k, the set $I - I_k$ can be covered by $2n$ cells, each one having one side $1/k$. So, (using sub-additivity of m^*), $m^*(I - I_k)$ is bounded above by $2n\frac{L^n}{k}$, where L is the maximum side length of I. Thus, $\lim_{k\to\infty} m^*(I - I_k) = 0$.

For each k we have, $A \cap I = (A \cap I_k) \cup (A \cap (I - I_k))$ and $A \cap I_k \subseteq A \cap I$. Hence

$$m^*(A \cap I_k) \leq m^*(A \cap I) \leq m^*(A \cap I_k) + m^*(A \cap (I - I_k)) \leq m^*(A \cap I_k) + m^*(I - I_k)$$

Taking limit as $k \to \infty$, we have $m^*(A \cap I) = \lim_{k\to\infty} m^*(A \cap I_k)$.

Now recall that the outer measure is additive if the sets in consideration have positive distance (Theorem 3.19). For each $k \geq 1$, $\text{dist}(A \cap I_k, A \cap I^c) \geq 1/k$. Also,for each $k \geq 1$, $(A \cap I_k) \cup (A \cap I^c) \subseteq A$. Hence we have

$$m^*(A) \geq m^*\left((A \cap I_k) \cup (A \cap I^c)\right) = m^*(A \cap I_k) + m^*(A \cap I^c).$$

Taking limit as $k \to \infty$ we obtain the desired inequality:

$$m^*(A) \geq m^*(A \cap I) + m^*(A \cap I^c)$$

□

Lemma 3.31. *Every open subset of $\mathbb{R}^n$ is a countable union of open cells.*

Proof. For an element $x \in \mathbb{R}^n$ and $r > 0$, let $C(x, r)$ be the open "cube" whose center is x and has side length r. That is, $C(x, r) = (x - \frac{r}{2}, x + \frac{r}{2}) \times (x - \frac{r}{2}, x + \frac{r}{2}) \times \cdots \times (x - \frac{r}{2}, x + \frac{r}{2})$.

Let $\mathbb{Q}^n = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid x_i \in \mathbb{Q})\}$ and note that $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$, that is, every open subset of $\mathbb{R}^n$ contains a point from $\mathbb{Q}^n$.

Let $U \subseteq \mathbb{R}^n$ be an open set. Since the set $U \cap \mathbb{Q}^n$ is countable we can list its elements: i.e., let $U \cap \mathbb{Q}^n = \{u_1, u_2, \ldots\}$.

For each $k \geq 1$, choose an integer $n_k > 0$ such that $C(u_k, \frac{1}{n_k}) \subseteq U$ and n_k is the smallest such positive integer. (Such n_k exists since U is an open set).

Let $O = \bigcup_{k=1}^{\infty} C(u_k, \frac{1}{n_k})$, so that O is a countable union of open cells.

We claim that $O = U$. Clearly, $O \subseteq U$. Conversely, let $u \in U$. Let m be a positive integer such that $C(u, \frac{1}{m}) \subseteq U$. By density of $\mathbb{Q}^n$, the open cell $C(u, \frac{1}{2m}) \subseteq U$ contains a point from $\mathbb{Q}^n$. Let this point be u_k. Recall that n_k is the smallest positive integer such that $C(u_k, \frac{1}{n_k}) \subseteq U$. Since $u_k \in C(u, \frac{1}{2m})$ we have that $C(u_k, \frac{1}{2m}) \subseteq C(u, \frac{1}{n}) \subseteq U$. Thus, we must have $n_k \leq 2m$ i.e., $\frac{1}{2m} \leq \frac{1}{n_k}$. But then we have $u \in C(u_k, \frac{1}{n_k}) \subseteq O$. □

Corollary 3.32. *The Borel σ-algebra $\mathcal{B}(\mathbb{R}^n)$ is contained in the Lebesgue σ-algebra $\mathcal{L}(\mathbb{R}^n)$.*

Proof. Recall that $\mathcal{B}(\mathbb{R}^n)$ is the σ-algebra generated by the open subsets of $\mathbb{R}^n$. That is, $\mathcal{B}(\mathbb{R}^n)$ is the smallest σ-algebra containing the open sets. The previous Lemma implies that all open sets of $\mathbb{R}^n$ are contained in $\mathcal{L}(\mathbb{R}^n)$ and hence $\mathcal{B}(\mathbb{R}^n)$ is contained in $\mathcal{L}(\mathbb{R}^n)$. □

We will later see that the collection $\mathcal{L}(\mathbb{R}^n)$ is actually much larger than $\mathcal{B}(\mathbb{R}^n)$.

The next theorem establishes the fact that m is translation invariant.

Theorem 3.33. *If $E \in \mathcal{L}(\mathbb{R}^n)$ and $x \in \mathbb{R}^n$ then $x + E \in \mathcal{L}(\mathbb{R}^n)$ and $m(E) = m(x + E)$.*

Proof. Let $A \subseteq \mathbb{R}^n$. Note that for any $y \in \mathbb{R}^n$ and $B \subseteq \mathbb{R}$, $(y + A) \cap B = y + (A \cap (-y + B))$. Since we have already shown m^* is translation invariant (Theorem 3.22), we have

$$m^*((y + A) \cap B) = m^*(A \cap (-y + B)) \text{ for any } y \in \mathbb{R}^n, B \subseteq \mathbb{R}^n$$

Now suppose E is measurable and $x \in \mathbb{R}^n$. For an arbitrary $A \subseteq \mathbb{R}^n$ we have

$$m^*(A) = m^*(-x + A) = m^*((-x + A) \cap E) + m^*((-x + A) \cap E^c).$$

Also, $m^*((-x + A) \cap E) = m^*(A \cap (x + E))$ (above equation with $y = -x, B = E$) and $m^*((-x + A) \cap E^c) = m^*(A \cap (x + E^c)) = m^*(A \cap (x + E)^c)$. Hence,

$$m^*(A) = m^*(A \cap (x + E)) + m^*(A \cap (x + E)^c)$$

and thus $x + E$ is measurable. Therefore $m(x + E) = m^*(x + E) = m^*(E) = m(E)$. □

We have defined a measure m on the σ-algebra $\mathcal{L}(\mathbb{R}^n)$ which contains all cells and m extends the usual notion of measure of cells. The next theorem shows that this extension is unique.

Theorem 3.34. *Let μ be a measure defined on $\mathcal{L}(\mathbb{R}^n)$ such that for any cell $I \subseteq \mathbb{R}^n$, $\mu(I) = \ell(I)$. Then $\mu = m$.*

Proof. If $E \in \mathcal{L}(\mathbb{R}^n)$ and $E \subseteq \bigcup_{k=1}^{\infty} I_k$ for some cells (I_k) then,

$$\mu(E) \leq \mu(\bigcup_{k=1}^{\infty} I_k) \leq \sum_{k=1}^{\infty} \mu(I_k) = \sum_{k=1}^{\infty} \ell(I_k).$$

Hence, $\mu(E) \leq m^*(E) = m(E)$ for any $E \in \mathcal{L}(\mathbb{R}^n)$.

To see the converse inequality, first assume that $E \in \mathcal{L}(\mathbb{R}^n)$ is bounded, i.e., there is a cell I such that $E \subseteq I$. Then, $m(E) + m(I - E) = m(I) = \mu(I) = \mu(E) + \mu(I - E)$. But we observed that $\mu(I - E) \leq m(I - E)$, hence $m(E) = \mu(E) + \mu(I - E) - m(I - E) \leq \mu(E)$. Thus μ and m agree on bounded measurable sets.

Now let $E \in \mathcal{L}(\mathbb{R}^n)$ be arbitrary and for each $k \geq 1$ let $I_k = (-k, k) \times (-k, k) \times \cdots \times (-k, k)$. Letting $E_1 = E \cap I_1$ and $E_k = E \cap (I_k - I_{k-1})$ for $k > 1$, we have (since E_k are disjoint, bounded measurable sets),

$$\mu(E) = \mu(\bigcup_{k=}^{\infty} E_k) = \sum_{k=1}^{\infty} \mu(E_k) = \sum_{k=1}^{\infty} m(E_k) = m(E)$$

□

Here is a characterization of measurable set in terms of open sets:

Theorem 3.35. *A subset $E \subseteq \mathbb{R}^n$ is measurable if and only if for every $\epsilon > 0$, there is an open set U such that $E \subseteq U$ and $m^*(U - E) \leq \epsilon$.*

Proof. Let $\epsilon > 0$ and U be an open subset such that $E \subseteq U$ and $m^*(U - E) \leq \epsilon$. Given a subset $A \subseteq \mathbb{R}^n$, We have, $A \cap E \subseteq A \cap U$ and $A - E = A \cap (U - E) \cup (A - U) \subseteq (U - E) \cup (A - U)$. So,

$$m^*(A \cap E) + m^*(A - E) \leq m^*(A \cap U) + m^*(U - E) + m^*(A - U).$$

Since U is open it is measurable, hence $m^*(A) = m^*(A \cap U) + m^*(A - U)$. Thus, $m^*(A \cap E) + m^*(A - E) \leq m^*(A) + \epsilon$. Since ϵ is arbitrary, we have $m^*(A \cap E) + m^*(A - E) \leq m^*(A)$, which shows E is measurable by Lemma 3.27.

Conversely, suppose that E is measurable and let $\epsilon > 0$. If $\mu^*(E) < \infty$, let (I_k) be a covering of E by open cells such that $\sum_{k=1}^{\infty} \ell(I_k) \leq m^*(E) + \epsilon$. Let $U = \bigcup_{k=1}^{\infty} I_k$ so that U is open and contains E. Since both U and E are measurable and $m^*(E) = m(E) < \infty$, we have $m^*(U - E) = m(U - E) = m(U) - m(E) = m^*(U) - m^*(E) \leq \sum_{k=1}^{\infty} \ell(I_k) + m^*(E) \leq \epsilon$.

Now let E be an arbitrary measurable set and let $E_k = E \cap I_k$ where I_k is the open cell centered at the origin with side length k. Since $m^*(E_k) < \infty$, for each k let U_k be an open set containing

E_k such that $m^*(U_k - E_k) \leq \frac{\epsilon}{2^k}$. Let $U = \bigcup_{k=1}^{\infty} U_k$. Then

$$m^*(U - E) \leq m^*\left(\bigcup_{k=1}^{\infty} U_k - E_k\right) \leq \sum_{k=1}^{\infty} m^*(U_k - E_k) \leq \epsilon$$

□

Proposition 3.36. *Let $A \subseteq \mathbb{R}^n$. Then $m^*(A) = \inf\{m(U) \mid U \text{ is open and } A \subseteq U\}$*

Proof. Let $\lambda(A) = \inf\{m(U) \mid U \text{ is open and } A \subseteq U\}$. Since we can replace cells by open cells in the definition of m^*, we have $\lambda(A) \leq m^*(A)$. Also, if U is open with $A \subseteq U$, then we have $m^*(A) \leq m^*(U) = m(U)$. Hence $m^*(A) \leq \lambda(A)$. □

The next theorem is the analogue of Theorem 3.35 for closed sets:

Theorem 3.37. *A subset $E \subseteq \mathbb{R}^n$ is measurable if and only if for every $\epsilon > 0$, there is a closed set F such that $F \subseteq E$ and $m^*(E - F) \leq \epsilon$.*

Proof. Suppose E is measurable. Given $\epsilon > 0$, since E^c is measurable, using Theorem 3.35, there is an open set G such that $E^c \subseteq G$ and $m(G - E^c) < \epsilon$. Let $F = G^c$ so that $m(E - F) = m(G - E^c) < \epsilon$.

Conversely, Let $A \subseteq \mathbb{R}^n$. To show E is measurable, we will show that $m^*(A) \geq m^*(A \cap E) + m^*(A \cap E^c)$. Let $\epsilon > 0$ and let F be a closed set such that $F \subseteq E$ $m^*(E - F) < \epsilon$. Since F is closed it is measurable and hence $m^*(A) = m^*(A \cap F) + m^*(A \cap F^c)$. Since $A \cap E = (A \cap F) \cup (A \cap (E - F))$ we have

$$\begin{aligned} m^*(A \cap E) + m^*(A \cap E^c) &\leq m^*(A \cap F) + m^*(A \cap (E - F)) + m^*(A \cap E^c) \\ &\leq m^*(A \cap F) + \epsilon + m^*(A \cap F^c) = m^*(A) + \epsilon \end{aligned}$$

Now the result follows since ϵ is arbitrary. □

As a corollary, we will show that the Lebesgue measure of a set can be approximated from within by measures of closed subsets:

Corollary 3.38. *Let $E \subseteq \mathbb{R}^n$ be measurable. Then*

$$m(E) = \sup\{m(F) \mid F \text{ is closed and } F \subseteq E\}$$

Proof. Let $\lambda(E) = \sup\{m(F) \mid F \text{ is closed and } F \subseteq E\}$. If $F \subseteq E$ is closed then $m(F) \leq m(E)$ and hence $\lambda(E) \leq m(E)$. Conversely, given $\epsilon > 0$, using the previous theorem, let $F \subseteq E$ be closed with $m(E - F) \leq \epsilon$. Then $m(E) = m(F) + m(E - F) = m(F) + \epsilon \leq \lambda(E) + \epsilon$. Hence $m(E) \leq \lambda(E)$. □

The next two results show that measurable subsets can be approximated from within by compact sets. Note that compact subsets have finite Lebesgue measure (exercise).

Theorem 3.39. *Let $E \subseteq \mathbb{R}^n$ have finite outer measure. Then E is measurable if and only if for every $\epsilon > 0$, there is a compact set C such that $C \subseteq E$ and $m^*(E - C) \leq \epsilon$.*

Proof. Since compact subsets of $\mathbb{R}^n$ are closed, one direction follows from Theorem 3.37. For the converse, suppose E is measurable. Let $B(k)$ denote the closed ball of radius k in $\mathbb{R}^n$ centered at the origin and let $E_k = E \cap B(k)$. Then (E_k) is an increasing sequence with union E. Hence $(m(E_k))$ converges to $m(E) < \infty$. Given $\epsilon > 0$ let $k_0 \geq 0$ such that $m(E) < m(E_{k_0}) + \epsilon/2$. Also (using Theorem 3.37) let $C \subseteq E_{k_0}$ be a closed subset such that $m(E_{k_0} - C) < \epsilon/2$. Note that C is compact since it is closed and bounded. We have,

$$m(E - C) = m(E - E_{k_0}) + m(E_{k_0} - C) < \epsilon/2 + \epsilon/2 = \epsilon$$

□

Corollary 3.40. *Let $E \subseteq \mathbb{R}^n$ be measurable. Then*

$$m(E) = \sup\{m(C) \mid C \text{ is compact and } C \subseteq E\}$$

Proof. If E has finite measure, the equality follows from the previous theorem and an argument similar to the proof of Corollary 3.38.(exercise)

If $m(E) = \infty$, $B(k)$ denote the closed ball of radius k in $\mathbb{R}^n$ centered at the origin and let $E_k = E \cap B(k)$. Hence $(m(E_k))$ converges to $m(E) = \infty$. Since E_k has finite measure, from the finite case we have for each $k \geq 1$ we have $m(E_k) = \sup\{m(C) \mid C \text{ is compact and } C \subseteq E_k\}$. Let $C_k \subseteq E_k \subseteq E$ be a compact subset such that $m(E_k) - 1 \leq m(C_k)$. Clearly $(m(C_k))$ converges to ∞. Then, $\sup\{m(C) \mid C \text{ is compact and } C \subseteq E\} \geq m(C_k)$ for every $k \geq 1$ and hence $\sup\{m(C) \mid C \text{ is compact and } C \subseteq E\} = \infty = m(E)$. □

3.6. **Null Sets.**

The following notion is fundamental in measure theory.

Definition 3.41. A subset $E \subseteq \mathbb{R}^n$ is a *null set* (or *a set of measure zero*) if $m^*(E) = 0$.

Exercises 3.42. Show that every countable subset of $\mathbb{R}^n$ is a null set.

Theorem 3.43. *A null set is measurable. Thus, every subset of a null set is measurable.*[1]

Proof. Suppose $m^*(E) = 0$. Let $A \subseteq \mathbb{R}^n$ be arbitrary. Then, (since $m^*(A \cap E) \leq m^*(E) = 0$), $m^*(A) \geq m^*(A \cap E^c) = m^*(A \cap E^c) + m^*(A \cap E)$ hence E is measurable.

If $F \subseteq E$ and $m^*(E) = 0$, then $m^*(F) = 0$ and hence F is measurable. □

So, the σ-algebra $\mathcal{L}(\mathbb{R}^n)$ contains all null sets. Next, we will show that there exists an uncountable null set in $\mathbb{R}$. This is the famous (middle thirds) *Cantor Set*. The Cantor set will be formed by starting with the interval $[0, 1]$ removing the "middle thirds" $(\frac{1}{3}, \frac{2}{3})$ and then iteratively performing this removal of middle thirds on any remaining piece. Here is a formal definition:

[1]Measures for which subsets of null sets are measurable are called *complete*. Thus the Lebesgue measure is complete. We may talk about this in later sections.

Definition 3.44. Let $C_0 = [0,1]$

$$C_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right] = C_0 - \bigcup_{i=0}^{3^0-1} \left(\frac{3i+1}{3}, \frac{3i+2}{3}\right)$$

$$C_2 = \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right] = C_1 - \bigcup_{i=0}^{3^1-1} \left(\frac{3i+1}{3^2}, \frac{3i+2}{3^2}\right)$$

$$\vdots$$

$$C_n = C_{n-1} - \bigcup_{i=0}^{3^{n-1}-1} \left(\frac{3i+1}{3^n}, \frac{3i+2}{3^n}\right)$$

The (middle thirds) Cantor Set is $C = \bigcap_{i=0}^{\infty} C_i$.

Since C and each C_n are closed and hence measurable. Note that $m(C_n) = (\frac{2}{3})^n$ and hence $m(C) = 0$ i.e., C is a null set.

Here is another description of C:

Every real number in $[0,1]$ has a base 3 expansion of the form,

$$0.a_1a_2a_3\ldots = \sum_{i=1}^{\infty} \frac{a_i}{3^i} \text{ where } a_i \in \{0,1,2\}.$$

This representation is unique except for rational numbers of the form $\dfrac{p}{3^k}$ $(p, k \in \mathbb{N}^+)$ there are two such expressions. For example $\dfrac{2}{3} = 0.2000\cdots = 0.122222\cdots$. Numbers of the form $\dfrac{p}{3^k}$ $p \leq 3^k, 3 \nmid p$ are exactly the ones for which $\exists k, a_i = 0, \forall i \geq k$. For such numbers, one representation has $a_i = 0$ $\forall i > k$ and one with $a_i = 2$ $\forall i > k$. In one case $a_k \in \{0,2\}$ in the other $a_k = 1$. Let us agree that for such numbers we will use the representation for which $a_k \in \{0,2\}$.

For a real number with base 3 expansion $r = 0.a_1a_2a_3\cdots$,

$$a_1 = 1 \iff \frac{1}{3} < r < \frac{2}{3} \iff r \notin C_1$$

$$a_1 \neq 1 \text{ and } a_2 = 1 \iff \frac{1}{9} < r < \frac{2}{9} \text{ or } \frac{7}{9} < r < \frac{8}{9} \iff r \notin C_2$$

$$\vdots$$

Thus, C consists of all real numbers $r \in [0,1]$ for which the base 3 expansion is of the form $r = 0.a_1a_2a_3\cdots$ with $a_i \neq 1, \forall i \geq 1$.

Now consider the function $f : C \to [0,1]$ defined as follows:

$$f\left(\sum_{i=1}^{\infty} \frac{a_i}{3^i}\right) = \sum_{i=1}^{\infty} \frac{a_i/2}{2^i}$$

Since every number in $[0,1]$ has a base two expansion, it follows that f is onto. Hence, C is an uncountable set.

Thus, we have proven the following.

Theorem 3.45. *The Cantor Set C is an uncountable null set of $\mathbb{R}$.*

One can modify this construction and obtain similar sets in $\mathbb{R}^n$. As a corollary of this fact we have:

Corollary 3.46. *The Lebesgue σ-algebra $\mathcal{L}(\mathbb{R}^n)$ has the same cardinality as $\mathcal{P}(\mathbb{R}^n)$.*

Proof. Since $\mathcal{L}(\mathbb{R}^n) \subseteq \mathcal{P}(\mathbb{R}^n)$, the cardinality of $\mathcal{L}(\mathbb{R}^n)$ is at most the cardinality of $\mathcal{P}(\mathbb{R}^n)$. Let C be an uncountable null set in $\mathbb{R}^n$. Since C is a null set, every subset of C is measurable. So, $\mathcal{L}(\mathbb{R}^n)$ contains all sets in $\mathcal{P}(C)$ which has the same cardinality as $\mathcal{P}(\mathbb{R}^n)$. So, $\mathcal{L}(\mathbb{R}^n)$ has the same cardinality as $\mathcal{P}(\mathbb{R}^n)$.

□

To complete this section, let us mention the following theorem without a proof:

Theorem 3.47. *The Borel σ-algebra $\mathcal{B}(\mathbb{R}^n)$ has the same cardinality as $\mathbb{R}$.*

The next theorem shows the relationship of $\mathcal{L}(\mathbb{R}^n)$ with $\mathcal{B}(\mathbb{R}^n)$:

Theorem 3.48. $\mathcal{L}(\mathbb{R}^n) = \{B \cup N \mid B \in \mathcal{B}(\mathbb{R}^n)$ *and* N *is a null set*$\}$

Proof. Let $E \in \mathcal{L}(\mathbb{R}^n)$. Using Theorem 3.35, for each $k \geq 1$ let U_k be an open set such that $E \subseteq U_k$ and $m(U_k - E) \leq \frac{1}{k}$. Let $B = \bigcap_{k=1}^{\infty} \in \mathcal{B}(\mathbb{R}^n)$, then $E \subseteq B$ and $m(B - E) = m(\bigcap_{k=1}^{\infty} U_k - E) \leq m(U_k - E) \leq \frac{1}{k}$. Thus $m(B - E) = 0$. Hence, for every $E \in \mathcal{L}(\mathbb{R}^n)$ there is $B \in \mathcal{B}(\mathbb{R}^n)$ such that $E \subseteq B$ and $m(B - E) = 0$.

Now let $E \in \mathcal{L}(\mathbb{R}^n)$. Since $E^c \in \mathcal{L}(\mathbb{R}^n)$. using the observation of the previous paragraph, There is $B \in \mathcal{B}(\mathbb{R}^n)$ such that $E^c \subseteq B$ and $m(B - E^c) = 0$. Since $B - E^c = E - B^c$ and $B^c \subseteq E$, we have $E = B^c \cup (E - B^c) = B^c \cup (B - E^c)$. Clearly, $B^c \in \mathcal{B}(\mathbb{R}^n)$ and $B - E^c$ is a null set. □

Exercises 3.49.

(1) Let $U \subseteq \mathbb{R}^n$ be an open set. Show that $m(U) > 0$ if and only if $U \neq \varnothing$.
(2) Let $K \subseteq \mathbb{R}^n$ be a compact subset. Show that $m(K) \neq \infty$.

4. Generation of measures

Recall that we have constructed the Lebesgue measure by starting with measure of simple sets. In this section we will see a systematic way of constructing measures on spaces starting with measures of simple sets. This can be viewed as an alternative (but equivalent) way of constructing the Lebesgue measure. The abstract versions of results will be used to construct product measures in further sections. Some results of this section will not be proved since their proof is analogous to similar statements of Section 3.

We will start with extending measures from algebras to σ-algebras.

Definition 4.1. Let X be a set and $\mathcal{A}$ be a family of subsets of X. $\mathcal{A}$ is an *algebra* if

(1) $\varnothing \in \mathcal{A}$,
(2) If $A \in \mathcal{A}$ then $A^c = X - A \in \mathcal{A}$
(3) If $A_1, \ldots, A_n \in \mathcal{A}$, then $\bigcup_{i=1}^n A_i \in \mathcal{A}$. (i.e., $\mathcal{A}$ is closed under finite unions)

Let $\mathcal{E}_L$ be the collection all subsets of $\mathbb{R}$ of the form $(a, b], (a, \infty), -\infty \leq a \leq b < \infty$ (i.e., $\mathcal{E}_L$ is the collection of all left open, right closed intervals). Then $\mathcal{E}_L$ satisfies the hypotheses of the next proposition (exercise).[1]

Similarly, all set s in $\mathbb{R}^n$ which are products of sets of the form $(a, b], (a, \infty), -\infty \leq a \leq b < \infty$ satisfy the hypotheses of the next proposition.

Proposition 4.2. *Let X be a set and $\mathcal{E} \subseteq \mathcal{P}(X)$ such that* [2]

(1) $\varnothing \in \mathcal{E}$,
(2) *$\mathcal{E}$ is closed under finite intersections,*
(3) *If $E \in \mathcal{E}$, then E^c is a finite disjoint union of members of $\mathcal{E}$.*

Let $\mathcal{A}$ be the collection of sets which finite disjoint unions of sets in $\mathcal{E}$, that is

$$\mathcal{A} = \left\{ A \subseteq X \mid A = \bigsqcup_{i=1}^n E_i, \text{ where } E_1, \ldots, E_n \in \mathcal{E} \text{ are disjoint} \right\}$$

Then $\mathcal{A}$ is an algebra containing $\mathcal{E}$.

Proof. Clearly $\varnothing \in \mathcal{A}$. Let $A = \bigsqcup_{i=1}^n E_i \in \mathcal{A}$ where $E_1, \ldots, E_n \in \mathcal{E}$ are disjoint. For each $i = 1, \ldots, n$ let $E_i^c = \bigsqcup_{j=1}^{m_i} B_j^i$ where $B_1^i, \ldots, B_{m_i}^i \in \mathcal{E}$ are disjoint. By introducing the empty set if needed, we can assume without loss of generality that $m_1 = m_2 = \cdots = m_n = m$. Then $A^c = \bigcap_{i=1}^n \bigsqcup_{j=1}^m B_j^i = \bigsqcup_{j=1}^m \bigcap_{i=1}^n B_j^i \in \mathcal{A}$ and hence $\mathcal{A}$ is closed under complements.

Now, let $E, F \in \mathcal{E}$ and write $F^c = \bigsqcup_{i=1}^n C_i$ where $C_i \in \mathcal{E}$ are disjoint. Then $E - F = E \cap F^c = \bigsqcup_{i=1}^n E \cap C_i \in \mathcal{A}$ and $E \cup F = (E - F) \bigsqcup F \in \mathcal{A}$. It follows by an inductive argument that

[1] As you may guess, there is nothing special about left open intervals here. One could also develop the theory with right open intervals.

[2] Such a collection is sometimes called an *elementary family*.

if $E_1, E_2, \dots E_n \in \mathcal{E}$, then $\bigcup_{i=1}^n E_i \in \mathcal{A}$. Thus, if $A, B \in \mathcal{A}$ then $A \cup B \in \mathcal{A}$. As usual, an inductive argument now shows that $\mathcal{A}$ is closed under finite unions.

□

Let us denote the algebra on $\mathbb{R}$ corresponding to $\mathcal{E}_L$ by $\mathcal{A}_L$. In other words, A_L consists of subsets of $\mathbb{R}$ which are finite disjoint unions of intervals of the form $(a, b], (a, \infty), -\infty \leq a \leq b < \infty$.

Definition 4.3. Let X be a set and $\mathcal{A}$ an algebra on X. A *pre-measure* on X is a function $\mu : \mathcal{A} \to [0, \infty]$ satisfying

(1) $\mu(\varnothing) = 0$

(2) for any disjoint sequence $(A_n) \in \mathcal{A}$, if $\bigcup_{n=1}^\infty A_n \in \mathcal{A}$, we have $\mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$.

The next proposition shows that the length function defines a pre-measure on $\mathcal{A}_L$:

Proposition 4.4. *Let $A \in \mathcal{A}_L$. Define $\ell(A) \in [0, \infty]$ (the length of A) as follows: If $A = \bigsqcup I_i$ where $I_i \in \mathcal{E}_L$ and for some i, I_i is of the form (a, ∞) or $(-\infty, b]$, then $\ell(A) = \infty$. Otherwise, if $A = \bigsqcup_{i=1}^n (a_i, b_i]$, then $\ell(A) = \sum_{i=1}^n (b_i - a_i)$. Then ℓ defines a pre-measure on $\mathcal{A}_L$*

Proof. Firstly, we need to show that ℓ is well defined since a set could be represented as a finite disjoint union in many ways. It is clear that if one representation contains an interval of the form (a, ∞) or $(-\infty, b]$ then any other representation contains an interval of this form. Hence, both representations will have infinite length.

Suppose $(a_i, b_i], i = 1, \dots, n$ are disjoint and $\bigsqcup_{i=1}^n (a_i, b_i] = (a, b]$. Then, after relabeling the indices we have $a = a_1 < b_1 = a_2 < b_2 = a_3 < \cdots a_n < b_n = b$. Hence, $\sum_{i=1}^n (b_i - a_i) = b - a$.

Now suppose $\bigsqcup_{i=1}^n (a_i, b_i] = \bigsqcup_{j=1}^m (c_j, d_j]$. For each $i = 1, \dots, n$ we have

$$(a_i, b_i] = \bigsqcup_{j=1}^{m} (a_i, b_i] \cap (c_i, d_j]$$

Note that $(a_i, b_i] \cap (c_i, d_j]$ is an interval of the form $(a, b]$. So, from the previous paragraph, for each $i = 1, \dots, n$ we have

$$b_i - a_i = \sum_{j=1}^{m} \ell((a_i, b_i] \cap (c_i, d_j])$$

Similarly, for each $j = 1, \dots, m$ we have

$$d_j - c_j = \sum_{i=1}^{n} \ell((a_i, b_i] \cap (c_i, d_j])$$

Now it follows that

$$\ell(\bigsqcup_{i=1}^{n} (a_i, b_i]) = \sum_{i=1}^{n} (b_i - a_i) = \sum_{i=1}^{n} \sum_{j=1}^{m} \ell((a_i, b_i] \cap (c_i, d_j]) = \sum_{j=1}^{m} \sum_{i=1}^{n} \ell((a_i, b_i] \cap (c_i, d_j])$$

$$= \sum_{j=1}^{m}(d_j - c_j) = \ell(\bigsqcup_{j=1}^{m}(c_j, d_j])$$

Thus, ℓ is well defined.

Suppose that $\bigsqcup_{i=1}^{\infty}(a_i, b_i] = (a, b]$. Let $(a_1, b_1], (a_2, b_2], \ldots, (a_n, b_n]$ be a finite collection. By re-labeling the indices if necessary, we have $a \leq a_1 < b_1 \leq a_2 \cdots < b_{n-1} \leq a_n < b_n \leq b$ and hence

$$\sum_{i=1}^{n} \ell(a_i, b_i] = \sum_{i=1}^{n}(b_i - a_i) \leq b_n - a_1 \leq b - a = \ell(a, b]$$

Since n is arbitrary, we have $\sum_{i=1}^{\infty} \ell(a_i, b_i] \leq \ell(a, b]$. Conversely, given $\epsilon > 0$, let (ϵ_i) be a sequence of positive real numbers such that $\sum \epsilon_i < \epsilon/2$. Consider the intervals $J_i = (a_i - \epsilon_i, b_i + \epsilon_i)$. Since $\bigsqcup_{i=1}^{\infty}(a_i, b_i] = (a, b]$, (J_i) forms an open cover of the compact interval $[a, b]$. Let $I_1, \ldots, I_m$ be a finite sub-cover. By renumbering and discarding some intervals we may assume that

$$a_1 - \epsilon_1 < a \ , \ \ b < b_m + \epsilon_m$$
$$a_i - \epsilon_i < b_{i-1} + \epsilon_i \ , \ \forall i = 2, 3, \ldots, m$$

So,

$$b - a \leq (b_m + \epsilon_m) - (a_1 - \epsilon_1) \leq \sum_{i=1}^{m}(b_i + \epsilon_i) - (a_i - \epsilon_i) \leq \sum_{i=1}^{\infty}(b_i - a_i) + 2\epsilon_i \leq \sum_{i=1}^{\infty}(b_i - a_i) + \epsilon$$

Since ϵ is arbitrary, we have $\ell(a, b] \leq \sum_{i=1}^{\infty} \ell(a_i, b_i]$. Thus if $\bigsqcup_{i=1}^{\infty}(a_i, b_i] = (a, b]$ then $\ell(a, b] = \sum_{i=1}^{\infty} \ell(a_i, b_i]$.

Now let (E_i) be a disjoint sequence in $\mathcal{A}_L$ with $\bigcup_{i=1}^{\infty} E_i \in \mathcal{A}$. Our aim is to show that $\ell(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \ell(E_i)$. If one of the sets E_i has an infinite interval this equality holds clearly. So assume that none of the sets E_i has an infinite interval in its decomposition. Without loss of generality we may assume that each $E_i = (a_i, b_i]$ for some $a_i \leq b_i$. Since $\bigcup_{i=1}^{\infty}(a_i, b_i] \in \mathcal{A}$, we have $\bigcup_{i=1}^{\infty}(a_i, b_i] = \bigcup_{j=1}^{m}(c_i, d_i]$ for some $c_i \leq d_i$.

Then, using the result of the previous paragraph we have:

$$\ell(\bigcup_{i=1}^{\infty}(a_i, b_i]) = \ell(\bigcup_{j=1}^{m}(c_i, d_i]) = \sum_{j=1}^{m} \ell(c_j, d_j] = \sum_{j=1}^{m}\sum_{i=1}^{\infty} \ell(c_j, d_j] \cap (a_i, b_i] = \sum_{i=1}^{\infty}\sum_{j=1}^{m} \ell(c_j, d_j] \cap (a_i, b_i]$$
$$= \sum_{i=1}^{\infty} \ell(a_i, b_i]$$

□

The next definition is analogous to the Lebesgue outer measure m^*.

Definition 4.5. Let $\mathcal{A}$ be an algebra on X and let μ be a pre-measure on $\mathcal{A}$. The function $\mu^* : \mathcal{P}(X) \to [0, \infty]$ defined by

$$\mu^*(E) = \inf\left\{\sum_{i=1}^{\infty} \mu(E_i) \mid (E_i) \in \mathcal{A}, E \subseteq \bigcup_{i=1}^{\infty} E_i\right\}$$

is called the *outer measure* generated by μ.

Theorem 4.6. *Let $\mathcal{A}$ be an algebra on X and let μ be a pre-measure on $\mathcal{A}$. Then*

(1) *If $E \subseteq F$ then $\mu^*(E) \leq \mu^*(F)$*
(2) *If $E \in \mathcal{A}$ then $\mu^*(E) = \mu(E)$*
(3) *If (E_i) is a sequence in $\mathcal{P}(X)$, then $\mu^*(\bigcup_{i=1}^{\infty} E_i) \leq \sum_{i=1}^{\infty} \mu^*(E_i)$*

Proof. These can be proven using similar arguments as the counterparts for m^*. □

Next we state the definition of measurability which is the exact analogue of the *Carathéodory condition* in the general setting.

Definition 4.7. Let X be a set, $\mathcal{A}$ an algebra on X and μ be a pre-measure on $\mathcal{A}$. A subset $E \subseteq X$ is *measurable* if

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c) \text{ for all subsets } A \subseteq X.$$

Theorem 4.8. *Let X be a set, $\mathcal{A}$ an algebra on X and μ be a pre-measure on $\mathcal{A}$. Let $\mathcal{A}^*$ be the collection of measurable subsets. $\mathcal{A}^*$ is a σ-algebra which contains $\mathcal{A}$. The restriction of μ to $\mathcal{A}^*$ is a measure and extends μ.*

Proof. The proof is analogous to the its counterpart for the Lebesgue outer measure. □

The next results show that under the assumption that μ is σ-finite (see Definition 3.10), this extension is unique.

Theorem 4.9. *(Extension Theorem[1]) Let X be a set, $\mathcal{A}$ an algebra on X and μ be a pre-measure on $\mathcal{A}$. Then,*

(1) *there is an extension of μ to a measure on $\mathcal{A}^*$.*

If μ is σ-finite

(2) *For any σ-algebra $\mathcal{B}$ with $\mathcal{A} \subseteq \mathcal{B} \subseteq \mathcal{A}^*$ any extension of μ to $\mathcal{B}$ is unique.*
(3) *In particular, there is a unique extension of μ to a measure on $\mathcal{A}^*$.*

Proof. The first statement is the previous theorem. Let $\mathcal{B}$ be a σ-algebra with $\mathcal{A} \subseteq \mathcal{B} \subseteq \mathcal{A}^*$ and let ν be a measure on $\mathcal{B}$ extending μ. Suppose first that $\mu(X) < \infty$. It follows that $\mu^*(X)$ and $\nu(X)$ are also finite. Let $E \in \mathcal{B}$ and let (E_n) be a sequence in $\mathcal{A}$ such that $E \subseteq \bigcup_{n=1}^{\infty} E_n$. Then

$$\nu(E) \leq \nu(\bigcup_{n=1}^{\infty} E_n) \leq \sum_{n=1}^{\infty} \nu(E_n) = \sum_{n=1}^{\infty} \mu(E_n)$$

Hence, $\nu(E) \leq \mu^*(E)$ for every $E \in \mathcal{B}$. Since

$$\nu(X) = \nu(E) + \nu(X - E) \leq \mu^*(E) + \mu^*(X - E) = \mu^*(X) = \nu(X) < \infty$$

[1]This theorem has various names attached to it in literature: Hahn, Carathedory, Frechet, Kolmogorov,...I was unable to pick one name

we have $\mu^*(E) = \nu(E)$ and hence the extension is unique.

Now if μ is σ-finite, let (F_n) be an increasing sequence in $\mathcal{A}$ such that $\mu(F_n) < \infty$ for every $n \geq 1$ and $X = \bigcup F_n$. From the first case, we have $\mu^*(E \cap F_n) = \nu(E \cap F_n)$ for every $E \in \mathcal{A}^1$ and $n \geq 1$. Thus for every $E \in \mathcal{B}$,

$$\mu^*(E) = \lim_{n\to\infty} \mu^*(E \cap F_n) = \lim_{n\to\infty} \nu(E \cap F_n) = \nu(E)$$

□

We now have another way of constructing the Lebesgue Measure on $\mathbb{R}$: The length function is a pre-measure on $\mathcal{A}_L$. The outer measure generated by the length function is clearly equal to the Lebesgue outer measure m^* (exercise).

4.1. **Measures on spaces of infinite sequences (This section is for the interested person and is not part of the lectures).**

Let $X = 2^{\mathbb{N}} = \{0,1\}^{\mathbb{N}}$ be the set of all infinite $0,1$ sequences. We write elements of X as $x = (x_n) = x_0x_1x_2\cdots$ where $x_i \in \{0,1\}$.

Given $x, y \in X$, with $x \neq y$, let $d(x,y) = \dfrac{1}{2^r}$ where $r = \min\{i \mid x_i \neq y_i\}$. Also define $d(x,x) = 0$ for every $x \in X$. As exercise, show that d gives a metric on X.

Given a finite sequence $x_0x_1\cdots x_n$ where $x_i \in \{0,1\}$, define

$$C(x_0x_1\cdots x_n) = \{y \in X \mid y_i = x_i, i = 0,1,\ldots,n\}$$

As exercise, show the following:

(1) For every $x \in C(x_0x_1\cdots x_n)$, $C(x_0x_1\cdots x_n)$ is the open ball of radius $\dfrac{1}{2^n}$ around x.
(2) For every $x \in C(x_0x_1\cdots x_n)$, $C(x_0x_1\cdots x_n)$ is the closed ball of radius $\dfrac{1}{2^{n+1}}$ around x.
(3) for $n \leq m$, if $C(x_0x_1\cdots x_n) \cap C(y_0y_1\cdots y_m) \neq \varnothing$, then $C(y_0y_1\cdots y_m) \subseteq C(x_0x_1\cdots x_n)$.

let $\mathcal{E}$ be the collection of subsets of X of the form $C(x_0x_1\cdots x_n)$ and also let $\varnothing \in \mathcal{E}$.

Show the following as exercise:

(1) $\mathcal{E}$ is closed under finite intersections,
(2) If $E \in \mathcal{E}$, then E^c is a finite disjoint union of members of $\mathcal{E}$.

Let $\mathcal{A}$ be the algebra corresponding to $\mathcal{E}$ given by Proposition 4.2. In other words, $\mathcal{A}$ is the algebra of finite disjoint unions of elements of $\mathcal{E}$.

For a set $C(x_0x_1\cdots x_n) \in \mathcal{E}$, let $\mu(C(x_0x_1\cdots x_n)) = \dfrac{1}{2^{n+1}}$ and $\mu(\varnothing) = 0$.

Define $\mu : \mathcal{A} \to [0,\infty]$ as follows: Given, $A \in \mathcal{A}$, write $A = \bigsqcup_{i=1}^k C_i$ where $C_i \in \mathcal{E}$. Then define $\mu(A) = \sum_{i=1}^k \mu(C_i)$.

As exercise, show that μ is well defined, gives a pre-measure on $\mathcal{A}$ and μ is σ-finite.

Now, by the extension theorem (Theorem 4.9), μ extends to a unique measure on $\mathcal{A}^*$.

Note that $\mu(X) = 1$ and hence μ could may serve as a model for a probability trial in which one tosses a fair coin infinitely many times.

Exercises 4.10.

(1) Suppose $\mathcal{E} \subseteq \mathcal{P}(X)$ satisfies
   (a) $\varnothing \in \mathcal{E}$,
   (b) $\mathcal{E}$ is closed under finite intersections,
   (c) If $E \in \mathcal{E}$, then E^c is a finite disjoint union of members of $\mathcal{E}$.

   Let $\mathcal{A} = \left\{ A \subseteq X \mid A = \bigsqcup_{i=1}^{n} E_i, \text{ where } E_1, \ldots, E_n \in \mathcal{E} \text{ are disjoint} \right\}$
   and $\mathcal{B} = \left\{ A \subseteq X \mid A = \bigcup_{i=1}^{n} E_i, \text{ where } E_1, \ldots, E_n \in \mathcal{E} \right\}$. Show that $\mathcal{A} = \mathcal{B}$.

(2) If $\mathcal{E}_L$ is the collection all subsets of $\mathbb{R}$ of the form $(a, b], (a, \infty), -\infty \leq a \leq b < \infty$, then $\mathcal{E}_L$ satisfies the conditions of Proposition 4.2. Also, show that the algebra $\mathcal{A}_L$ for this family is not a σ-algebra.

(3) Let X be a set, $\mathcal{Y} \subseteq \mathcal{P}(X)$ and let $\mathcal{A}_\mathcal{Y}$ be the sigma algebra generated by $\mathcal{Y}$. If $\mathcal{B}$ is a σ-algebra such that $\mathcal{Y} \subseteq \mathcal{B}$, show that $\mathcal{A} \subseteq \mathcal{B}$.

(4) The aim of this exercise is to show that if the pre-measure (as in the Extension Theorem) is not σ-finite, then the extension is not necessarily unique.

   Let $\mathcal{A}$ be the collection of finite unions of subsets of $\mathbb{Q}$ of the form $(a, b] \cap \mathbb{Q}$ where $-\infty \leq a < b \leq \infty$.
   (a) Show that $\mathcal{A}$ is an algebra.
   (b) Show that the σ-algebra generated by $\mathcal{A}$ is $\mathcal{P}(\mathbb{Q})$.
   (c) Define $\mu : \mathcal{A} \to [0, \infty]$ by

$$\mu(A) = \begin{cases} 0 & A = \varnothing \\ \infty & A \neq \varnothing \end{cases}$$

   Show that μ is a pre-measure.
   (d) Show that μ is NOT σ-finite.
   (e) For each $k \geq 0$ define $\mu_k : \mathcal{P}(\mathbb{Q}) \to [0, \infty]$ by

$$\mu_k(E) = \begin{cases} k \cdot |E| & \text{if } A \text{ is finite} \\ \infty & \text{otherwise} \end{cases}$$

   Show that μ_k is a measure.
   (f) Show that μ_k is an extension of μ

(5) Convince yourself about the statements after Theorem 4.9.

5. Measurable Functions

In this section we will investigate functions on a measure space which are called *measurable*. The definition given below will be for an abstract measure space X with a σ-algebra $\mathcal{A}$. For applications the space X will be mostly the set $\mathbb{R}^n$ with the Lebesgue or Borel σ-algebra. For a motivation, one should remember the characterization of continuous functions as the ones for which inverse images of open sets are open (See section 1.3.)

For a function $f : X \to \mathbb{R}$ to be measurable we want the inverse images of sets in the Borel σ-algebra of $\mathbb{R}$ to be measurable in X. The following condition ensures this:

Definition 5.1. Let X be a set and $\mathcal{A}$ be a σ-algebra on X. A function $f : X \to \mathbb{R}$ is *$\mathcal{A}$-measurable* (or just *measurable)* if for any $a \in \mathbb{R}$ the set

$$f^{-1}(a, \infty) = \{x \in X \mid f(x) > a\}$$

belongs to $\mathcal{A}$.

The next lemma shows equivalent conditions for measurability of a function:

Lemma 5.2. *Let $f : X \to \mathbb{R}$ be a function. The following are equivalent:*

1. *For any $a \in \mathbb{R}$, $f^{-1}(a, \infty) = \{x \in X \mid f(x) > a\} \in \mathcal{A}$*
2. *For any $a \in \mathbb{R}$, $f^{-1}(-\infty, a] = \{x \in X \mid f(x) \leq a\} \in \mathcal{A}$*
3. *For any $a \in \mathbb{R}$, $f^{-1}[a, \infty) = \{x \in X \mid f(x) \geq a\} \in \mathcal{A}$*
4. *For any $a \in \mathbb{R}$, $f^{-1}(-\infty, a) = \{x \in X \mid f(x) < a\} \in \mathcal{A}$*

Proof. We will only show that 1) implies 3), the rest being an exercise. Note that

$$[a, \infty) = \bigcap_{n=1}^{\infty} (a - \frac{1}{n}, \infty) \text{ and hence } f^{-1}[a, \infty) = \bigcap_{n=1}^{\infty} f^{-1}(a - \frac{1}{n}, \infty)$$

Thus, if 1) holds, 3) holds too. □

Example 5.3.

1. Let X be a set and $\mathcal{A}$ a σ-algebra. A constant function $f : X \to \mathbb{R}$ is $\mathcal{A}$-measurable. If $f(x) = c$ for any $x \in X$, then
$$f^{-1}(a, \infty) = \begin{cases} \varnothing & \text{if } c \leq a \\ X & \text{if } c > a \end{cases}$$
and $X, \varnothing$ belong to $\mathcal{A}$.
2. Let $E \in \mathcal{A}$. The *characteristic function* of E is the function $\chi_E : X \to \mathbb{R}$ defined by
$$\chi_E(x) = \begin{cases} 1 & \text{if } x \in E \\ 0 & \text{if } x \notin E \end{cases}$$
We have $\chi_E^{-1}(a, \infty) = \begin{cases} X & \text{if } a \leq 0 \\ E & \text{if } 0 < a < 1 \\ \varnothing & \text{if } a \geq 1 \end{cases}$, hence χ_E is $\mathcal{A}$-measurable.

(3) Consider $\mathbb{R}$ with the Borel σ-algebra $\mathcal{B}(\mathbb{R})$. If $f : \mathbb{R} \to \mathbb{R}$ is continuous, then for any $a \in \mathbb{R}$, $f^{-1}(a, \infty)$ is an open set in $\mathbb{R}$ and thus $f^{-1}(a, \infty) \in \mathcal{B}(\mathbb{R})$. Therefore, continuous functions $\mathbb{R} \to \mathbb{R}$ are $\mathcal{B}(\mathbb{R})$-measurable.

Note that since $\mathbb{Q} \in \mathcal{B}(\mathbb{R})$, the function $\chi_{\mathbb{Q}} : \mathbb{R} \to \mathbb{R}$ is $\mathcal{B}(\mathbb{R})$-measurable.

(4) Let $f : \mathbb{R} \to \mathbb{R}$ be an increasing function (i.e., if $x \leq y$ then $f(x) \leq f(y)$). Then for any $a \in \mathbb{R}$, $f^{-1}(a, \infty)$ is of the form $(b, \infty), [b, \infty)$ or one of $\mathbb{R}, \varnothing$ (Why?). Hence increasing functions are $\mathcal{B}(\mathbb{R})$-measurable. Similarly, decreasing functions are $\mathcal{B}(\mathbb{R})$-measurable.

Theorem 5.4. *Let X be a set and $\mathcal{A}$ be a σ-algebra on X. Suppose $f, g : X \to \mathbb{R}$ are measurable functions. Then the following functions are also measurable:*

(1) *For any $c \in \mathbb{R}$, cf*
(2) f^2
(3) $f + g$
(4) $f \cdot g$
(5) $|f|$

Proof.

(1) This is clear for $c = 0$. If $c > 0$, then

$$(cf)^{-1}(a, \infty) = \{x \in X \mid (cf)(x) > a\} = \{x \in X \mid f(x) > a/c\} = f^{-1}(a/c, \infty) \in \mathcal{A}.$$

Similarly, if $c < 0$, $(cf)^{-1}(a, \infty) = f^{-1}(-\infty, a/c) \in \mathcal{A}$ by Lemma 5.2.

(2) If $a \geq 0$, then

$$(f^2)^{-1}(a, \infty) = \{x \in X \mid f(x)^2 > a\} = \{x \in X \mid f(x) > \sqrt{a} \text{ or } f(x) < -\sqrt{a}\}$$
$$= f^{-1}(\sqrt{a}, \infty) \cup f^{-1}(-\infty, -\sqrt{a}) \in \mathcal{A}$$

If $a < 0$, then $(f^2)^{-1}(a, \infty) = \{x \in X \mid f(x)^2 > a\} = X \in \mathcal{A}$.

(3) We claim that for any $a \in \mathbb{R}$,

$$(f + g)^{-1}(a, \infty) = \bigcup_{r \in \mathbb{Q}} \left(f^{-1}(r, \infty) \cap g^{-1}(a - r, \infty) \right)$$

and clearly the right hand side belongs to $\mathcal{A}$ if f, g are measurable. To see this equality, note that $x \in (f + g)^{-1}(a, \infty) \iff f(x) + g(x) > a \iff f(x) > a - g(x) \iff f(x) > r > a - g(x)$ for some $r \in \mathbb{Q} \iff f(x) > r$ and $g(x) > a - r$ for some $r \in \mathbb{Q}$.

(4) Since $(f + g)^2 = f^2 + 2fg + g^2$, fg is measurable using the first three parts.

(5) If $a < 0$, then $|f|^{-1}(a, \infty) = \{x \in X \mid |f(x)| > a\} = X$. If $a \geq 0$, then $|f|^{-1}(a, \infty) = f^{-1}(a, \infty) \cup f^{-1}(-\infty, -a)$. In any case $|f|^{-1}(a, \infty) \in \mathcal{A}$.

□

Definition 5.5. Let X be a set and let $f : X \to \mathbb{R}$ be a function. Define the non-negative valued functions $f^+, f^- : X \to [0, \infty)$ by

$$f^+(x) = \max\{f(x), 0\}\ ,\ \forall x \in X \quad f^-(x) = \max\{-f(x), 0\}\ ,\ \forall x \in X.$$

f^+ is the *positive part* and f^- is the *negative part* of f.

Note the following relations:

$$f = f^+ - f^- \ , \ |f| = f^+ + f^- \ , \ f^+ = \frac{1}{2}(|f| + f) \ , \ f^- = \frac{1}{2}(|f| - f)$$

Thus, it follows that f is measurable if and only if both f^+ and f^- are measurable.

Occasionally, we want to consider functions $f : X \to \overline{\mathbb{R}}$. We define extended real valued measurable functions by:

Definition 5.6. Let X be a set and $\mathcal{A}$ be a σ-algebra on X. A function $f : X \to \overline{\mathbb{R}}$ is $\mathcal{A}$-*measurable* if for every $a \in \mathbb{R}$, $f^{-1}(a, \infty] = \{x \in X \mid f(x) > a\} \in \mathcal{A}$.

(Note that, as before, one can replace $(a, \infty]$ with $[a, \infty], [-\infty, a], [-\infty, a)$ in this definition. Also, if $f : X \to \mathbb{R}$ is a measurable function, then it is measurable when viewed as a function $f : X \to \overline{\mathbb{R}}$.)

For a function $f : X \to \overline{\mathbb{R}}$ we have

$$f^{-1}(\infty) = \{x \in X \mid f(x) = \infty\} = \bigcap_{k=1}^{\infty} f^{-1}(k, \infty]$$

and

$$f^{-1}(-\infty) = \{x \in X \mid f(x) = -\infty\} = \left(\bigcup_{k=1}^{\infty} f^{-1}(-k, \infty] \right)^c$$

So, if $f : X \to \overline{\mathbb{R}}$ is $\mathcal{A}$-measurable then $f^{-1}(\infty)$ and $f^{-1}(-\infty)$ belong to $\mathcal{A}$.

Lemma 5.7. *Let X be a set and $\mathcal{A}$ be a σ-algebra on X. A function $f : X \to \overline{\mathbb{R}}$ is measurable if and only if $f^{-1}(\infty), f^{-1}(-\infty)$ belong to $\mathcal{A}$ and the function $g_f : X \to \mathbb{R}$ defined by*

$$g_f(x) = \begin{cases} f(x) & \text{if } f(x) \neq \mp\infty \\ 0 & \text{otherwise} \end{cases}$$

is measurable.

Proof. Note that

$$g_f^{-1}(a, \infty) = \begin{cases} f^{-1}(a, \infty) = f^{-1}(a, \infty] - f^{-1}(\infty) & a \geq 0 \\ f^{-1}(a, \infty] \cup f^{-1}(-\infty) & a < 0 \end{cases}$$

Hence if f is measurable, then $f^{-1}(\infty), f^{-1}(-\infty)$ belong to $\mathcal{A}$ by observation before the lemma and hence g_f is measurable.

Similarly, note that

$$f^{-1}(a, \infty] = \begin{cases} g_f^{-1}(a, \infty) \cup f^{-1}(\infty) & a \geq 0 \\ g_f^{-1}(a, \infty) - f^{-1}(-\infty) & a < 0 \end{cases}$$

So, if $f^{-1}(\infty), f^{-1}(-\infty)$ belong to $\mathcal{A}$ and g_f is measurable, then f is measurable. □

Definition 5.8. Let X be a set and $\mathcal{A}$ be a σ-algebra on X. The collection of all $\mathcal{A}$-measurable functions $f : X \to \overline{\mathbb{R}}$ will be denoted by $M(X, \mathcal{A})$.

Lemma 5.9. *If $f \in M(X, \mathcal{A})$ then $cf, c \in \mathbb{R}, f^2, |f|, f^+, f^- \in M(X, \mathcal{A})$.*

Proof. These follow from Theorem 5.4 and the previous lemma. For example, (since $0 \cdot \mp\infty = 0$) we have $0 \cdot f = 0$. If $c \neq 0$ then $(cf)^{-1}(\mp\infty) = f^{-1}(\mp\infty)$ (depending on the sign of c) and $g_{cf} = cg_f$. So cf is measurable whenever f is.

$(f^2)^{-1}(\infty) = f^{-1}(\infty)$, $(f^2)^{-1}(-\infty) = \varnothing$ and $g_{f^2} = (g_f)^2$. Thus, if f is measurable then f^2 is measurable.

The rest is proven similarly.

□

Lemma 5.10. *Let (f_n) be a sequence of functions in $M(X, \mathcal{A})$ and define*

$$f(x) = \inf f_n(x) \quad , \quad F(x) = \sup f_n(x)$$
$$f^*(x) = \liminf f_n(x) \quad , \quad F^*(x) = \limsup f_n(x)$$

Then $f, F, f^*, F^* \in M(X, \mathcal{A})$.

Proof. Note that for a sequence (a_n) in $\overline{\mathbb{R}}$,

$$\inf a_n \geq a \iff \forall n, a_n \geq a$$
$$\sup a_n > a \iff \exists n, a_n > a$$

Hence we have $f^{-1}[a, \infty] = \bigcap_{n=1}^{\infty} f_n^{-1}[a, \infty]$ and $F^{-1}(a, \infty] = \bigcup_{n=1}^{\infty} f_n^{-1}(a, \infty]$. Thus f and F are measurable.

By the same reasoning, for any $n \geq 1$, the functions $\inf\{f_m(x) \mid m \geq n\}$ and $\sup\{f_m(x) \mid m \geq n\}$ are measurable. Since

$$f^*(x) = \sup_n \inf\{f_m(x) \mid m \geq n\} \text{ and } F^*(x) = \inf_n \sup\{f_m(x) \mid m \geq n\}$$

we conclude that f^* and F^* are measurable. □

Corollary 5.11. *Let (f_n) be a sequence of functions in $M(X, \mathcal{A})$. If (f_n) converges to f, then $f \in M(X, \mathcal{A})$.*

Proof. Note that in this case $f(x) = \lim f_n(x) = \liminf f_n(x)$. □

Lemma 5.12. *If $f, g \in M(X, \mathcal{A})$ then $f \cdot g \in M(X, \mathcal{A})$.*

Proof. For each $n \geq 1$ define $f_n : X \to \mathbb{R}$ by

$$f_n(x) = \begin{cases} f(x) & \text{if } -n \leq f(x) \leq n \\ n & \text{if } f(x) > n \\ -n & \text{if } f(x) < -n \end{cases}$$

Clearly (f_n) converges to f. As exercise show that f_n is measurable. Define g_n similarly.

Since f_n and g_n are real-valued functions, it follows from Theorem 5.4 that for any $n, m \geq 1$, $f_n \cdot g_m$ is measurable.

For each $m \geq 1$, $f(x)g_m(x) = \lim_{n\to\infty} f_n(x)g_m(x)$, hence by Corollary 5.11, for any $m \geq 1$, $f \cdot g_m \in M(X, \mathcal{A})$. Since $f(x)g(x) = \lim_{m\to\infty} f(x)g_m(x)$, again by Corollary 5.11, $f \cdot g \in M(X, \mathcal{A})$. □

Note that if $f, g \in M(X, \mathcal{A})$ then the sum $f + g$ is not well defined since we have not defined $\infty + (-\infty)$. If one defines $f + g$ to be zero whenever this occurs, the resulting function will be in $M(X, \mathcal{A})$.

The following theorem is of great interest.

Theorem 5.13. *Let f be a non-negative function in $M(X, \mathcal{A})$ (i.e., the image of f lies in $[0, \infty]$). Then there is a sequence (φ_n) of functions in $M(X, \mathcal{A})$ such that*

1. *$0 \leq \varphi_n(x) \leq \varphi_{n+1}(x) \leq f(x)$ for any $x \in X, n \geq 1$*
2. *$f(x) = \lim_{n\to\infty} \varphi_n(x)$ for any $x \in X$*
3. *For any $n \geq 1$, φ_n takes only finitely many values (i.e., the image of φ_n is a finite set).*

Proof. Fix $n \geq 1$ and consider the following partition of $[0, \infty]$ given by

$$\left[0, \frac{1}{2^n}\right), \left[\frac{1}{2^n}, \frac{2}{2^n}\right), \left[\frac{2}{2^n}, \frac{3}{2^n}\right), \cdots, \left[\frac{n2^n - 1}{2^n}, n\right), [n, \infty]$$

That is, the interval $[0, n)$ is divided into $n2^n$ half open intervals of equal lengths $\frac{1}{2^n}$. For $n \geq 1$ and $k \in \{0, 1, 2, \ldots, n2^n\}$ define $E_n^k \subseteq X$ by

$$E_n^k = \begin{cases} f^{-1}\left[\frac{k}{2^n}, \frac{k+1}{2^n}\right) & \text{if} \quad 0 \leq k < n2^n \\ f^{-1}[n, \infty] & \text{if} \quad k = n2^n \end{cases}$$

Clearly each E_n^k is measurable and each $n \geq 1$, $\{E_n^k \mid k = 0, 1, \ldots, n2^n\}$ is a partition of X.

Define $\varphi_n = \sum_{k=0}^{n2^n} \frac{k}{2^n} \chi_{E_n^k}$, i.e., $\varphi_n(x) = \frac{k}{2^n}$ if $x \in E_n^k$. (See the picture below which will be added later).

φ_n is a linear combination of characteristic functions of measurable sets, thus it is measurable. Properties (1) and (3) follow directly from the construction of φ_n. To see property (2), given $x \in X$, if $f(x) = \infty$, then $\varphi_n(x) = n$ for all n and hence $f(x) = \lim_{n\to\infty} \varphi_n(x)$. If $f(x) < \infty$, let N be the smallest integer such that $f(x) \leq N$. Then, for $n \geq N$ we have $0 \leq f(x) - \varphi_n(x) \leq \frac{1}{n}$. Thus, $f(x) = \lim_{n\to\infty} \varphi_n(x)$. □

Here is a more general definition of measurable functions between measure spaces.

Definition 5.14. Let X, Y be sets and $\mathcal{A}_X, \mathcal{A}_Y$ σ-algebras on X and Y respectively. A function $f : X \to Y$ is $\mathcal{A}_X, \mathcal{A}_Y$ measurable if for every $C \in \mathcal{A}_Y$, $f^{-1}(C) \in \mathcal{A}_X$.

Thus, in our initial definition, if X is a set and $\mathcal{A}$ a σ-algebra on X, then $f : X \to \mathbb{R}$ is measurable if it is $\mathcal{A}, \mathcal{B}(\mathbb{R})$ measurable.

Exercises 5.15.

1. Finish the proof of Lemma 5.2.

(2) Let $f \in M(X, \mathcal{A})$ and for each real number $r > 0$ define $f_r : X \to \mathbb{R}$ by

$$f_r(x) = \begin{cases} f(x) & \text{if} \quad -r \leq f(x) \leq r \\ r & \text{if} \quad f(x) > r \\ -r & \text{if} \quad f(x) < -r \end{cases}$$

Show that f_r is measurable.

(3) Give an example of function $f : X \to \mathbb{R}$ which is not measurable but $|f|, f^2$ are measurable.

(4) Convince yourself that φ_n in Theorem 5.13 satisfy conditions (1),(2),(3).

(5) Let $f : X \to [0, \infty]$ be a measurable function which is bounded (i.e., $\exists K \in \mathbb{R}$ such that $f(x) \leq K \ , \forall x$). Show that the sequence (φ_n) constructed in Theorem 5.13 converges to f uniformly, that is, for every $\epsilon > 0$ there is $N \geq 1$ such that $\forall n \geq N, \forall x \in X$, $|f(x) - \varphi_n(x)| < \epsilon$

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